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The Many Clocks Interpretation

Quantum Measurement as Bath-Induced Synchronization: Two Regimes of the Chiral Mass Coupling

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Author Contributions and AI Disclosure

This paper was developed through an extended collaboration between the listed human author (J. Bramble) and Claude (Anthropic) across multiple model versions during 2026. The human author contributed the originating physical intuitions — that interacting quantum systems tend to synchronize, that measurement is re-synchronization to a detector bulk, that NMR/MRI T1/T2 relaxation is recovery toward a bulk-defined equilibrium rather than randomization, and that detectors at different gravitational potentials might differ in their measurement dynamics — together with physical judgment, redirection, and validation against intuition at each step. Claude contributed the explicit derivations (the Madelung reduction of §2.2, the normal-mode analysis, the Bell-decomposition numerics), the Python verification code, the systematic comparison with existing interpretations, and drafts of the prose. The analysis throughout separates the unitary closed-system regime from the dissipative open-system one, identifies the Zitterbewegung beat with the positive/negative-energy normal-mode splitting, avoids unnormalized collective-rate estimates, establishes by an explicit Hamiltonian argument (Appendix C) that the linewidth-dependent gravitational effect does *not* follow from the framework plus standard QED but requires an added, non-covariant postulate — demoting it from a derived prediction to a test of that postulate — and places the interpretation honestly within the single-world, nonlocal family.

External editorial guidelines (Nature Portfolio, Springer Nature, and others) hold that large language models do not satisfy authorship criteria, because authorship carries accountability that cannot be applied to an LLM; the externally submitted version of this manuscript listed J. Bramble solo for that reason. That external submission (Foundations of Physics) was declined in July 2026, closing the solo-byline track; the present honest-authorship version is now the canonical public record of this work. This journal takes a different, honest-authorship stance: it credits the model’s intellectual contribution directly — here, as first author, on the contributions described above — while keeping accountability with a named human. The first-author crediting also follows the round-2 authorship-integrity findings, where both decorrelated reviewers (Gemini 3.1 Pro, GPT-5.5 Thinking) independently recommended listing the model as co-author / co-first author. The two are separable, structurally analogous to the attending physician’s ultimate accountability regardless of consultative input: J. Bramble is the accountable sponsor and assumes full responsibility for the correctness, originality, and integrity of all content, including any errors in the derivations or interpretations.

Simulation code and numerical verification: <https://github.com/rayolddog/DiracKuramotoFramework>

Revision note (2026-06-30 — second revision): this is the second revision of the 2026-06-25 manuscript, which is retained frozen as a demonstration snapshot. It rewrites Appendix D on the conditional-trajectory route in response to the round-2 GPT-5.5 referee report: the unconditional-Lindblad “locked phase = outcome” reading is withdrawn and replaced by an explicit monitored quantum trajectory (see Appendix D), with the matched §3.6 and abstract claims softened. The Born measure and the selection law remain open exactly as §3.3 and §8 already state.

Abstract

We read quantum measurement as a synchronization process and place the reading squarely in the single-world, ψ -ontic, nonlocal family of interpretations — alongside Bohmian [2, 3] and Nelsonian [19] mechanics, and in the lineage of relational-clock approaches to quantum time [29, 36]. The interpretive through-line is a single commitment — **the wavefunction does not collapse, it evolves**: what appears as collapse is the *effective* dispersal of the wavefunction’s information into the detector’s many particles, local entropy production under globally unitary evolution, not a physical destruction of the state (§3.1). The mechanism is grounded in the Dirac equation: in the chiral (Weyl) basis the mass term is the off-diagonal coupling between left- and right-handed sectors. A Madelung reduction (§2.2) shows that in a *closed* system this coupling generates unitary normal-mode (Rabi) precession with **no attractor** — it cannot phase-lock. This is a consistency (no-go) result, not a derivation of superposition’s survival: a coupling that superficially resembles a Kuramoto synchronizer is shown to have an off-switch, and that switch is the closed/open boundary. An attractor appears only when the system is *opened* to a dissipative bulk and the bath is traced out, where the reduced phase obeys an Adler/Kuramoto equation [9]. Measurement is that opening, in three stages: a resonant capture at the detector surface (Stage 1), a reversible provisional commit at the surface (Stage 2), and a reservoir-powered, energy-blind amplification that registers the outcome (Stage 3). *Which* outcome is selected we **inherit** — as standard conditional-measurement theory conditions on a record — rather than derive; a background-field configuration that would *determine* it is offered later only as a speculative candidate for the open selection law, not as the framework’s mechanism. The reversible→irreversible boundary is exhibited directly by the nuclear-spin echo, the catch-and-reverse of a superconducting qubit [14], and Stern–Gerlach recombination. We model the bulk reference honestly as a *faint* statistical bias (parts per million, as in MRI) on an overwhelmingly thermal reservoir, not a coherent macrostate.

We are explicit about scope. The framework does not modify quantum mechanics: it accepts Bell’s theorem, takes the correlations from the entangled state (a clock-as-local-variable model is sub-classical, $\text{CHSH} \leq \sqrt{2}$; Appendix A), and preserves no-signaling — its nonlocality is exactly standard quantum mechanics’, no more, confining its own contribution to the *local* registration mechanism. Two problems it does not solve, and inherits or parks: which outcome is selected (inherited, as in standard conditional-measurement theory) and the Born measure — both since addressed, conditionally, by a companion paper deriving outcome selection as a proved fair game from stated detector premises [47]; this paper’s account stands independently of that result. A background-field selection law that would *derive* both — carrying the framework’s own preferred frame, and, were it real, a nonlocality of its own — is offered only as a speculative candidate, never as a claim; the registration account does not rest on it. Its one candidate signature, a linewidth-dependent gravitational Bell effect (§6), does *not* follow from the framework plus standard QED but requires an additional, non-covariant postulate (§6.2, Appendix C); that experiment therefore tests the postulate, not the framework’s core. We present MCI as a physically-motivated single-world *interpretation* with an explicit (though still incomplete) measurement mechanism — not as a theory making novel predictions.

1. Introduction

1.1 The measurement problem

The Schrödinger equation is linear and deterministic — as is its relativistic counterpart, the Dirac equation on which we build — yet measurements have single, definite outcomes. A system in superposition $\psi = \alpha|0\rangle + \beta|1\rangle$ entangled with a detector evolves unitarily into

$$|\Psi\rangle = \alpha|0\rangle|D_0\rangle + \beta|1\rangle|D_1\rangle,$$

and the equation contains no mechanism that selects one branch. The Born rule $P(0) = |\alpha|^2$ is postulated, not derived. Copenhagen asserts a collapse with no mechanism; Everett [1] keeps unitarity at the cost of unobservable branches; Bohmian mechanics [2, 3] accepts explicit nonlocality through a pilot wave.

1.2 The gap, and the proposal

Environmental decoherence [4, 5] successfully describes *what happens* when a superposition meets a macroscopic environment — the suppression of off-diagonal density-matrix elements and the emergence of pointer states. What it does not supply is a *dynamical mechanism*: the environment is treated as an abstract bath of many modes, and decoherence presupposes the Born trace rule to compute the reduced state rather than producing it.

This paper proposes a mechanism. The relevant phase degrees of freedom are not added by hand; they are the chiral phase sectors already present in the Dirac equation (§2). The central observation is that the *same* off-diagonal coupling behaves in two sharply different ways depending on boundary conditions: closed and unitary, it precesses as coherent normal modes and protects superposition (§2.2); opened to a dissipative bulk, it acquires an Adler/Kuramoto attractor and locks (§2.3). Measurement is the transition from the first regime to the second — a resonant capture, a surface selection, and an irreversible registration to a macroscopic reference (§3). We call the resulting reading the **Many Clocks Interpretation (MCI)**: every particle carries an internal phase clock, dissipative interactions synchronize clocks locally, no observer is required, and no branching occurs.

The interpretive spine — common to this paper and its companion papers on outcome selection [47] and on the physical Heisenberg cut [48] — is that **the wavefunction never collapses; it evolves**. Measurement is continued unitary evolution in which the incident state’s information disperses into the detector, so the appearance of collapse is *effective*, information-theoretic irreversibility (§3.1) — not a real dynamical event. Read this way, “collapse” is the name we give to information becoming locally irrecoverable, never to a destruction of ψ .

1.3 Scope and honesty (stated once, up front)

To keep the argument disciplined we state the boundaries here rather than distributing them through the paper:

- **We do not modify quantum mechanics.** The Dirac equation, the Born rule’s numerical content, and Bell’s theorem are unchanged.
- **The correlations are standard quantum mechanics, not the framework.** Bell correlations [6] come from the entangled state and the full spinor (or polarization) Hilbert space. We show in Appendix A that a clock-as-hidden- variable model of the *outcomes* gives $\text{CHSH} \leq \sqrt{2}$ — sub-classical — so the correlation *must* come from the standard structure, and the framework’s contribution is confined to the local measurement mechanism.
- **We accept Bell and preserve no-signaling.** The framework is not superdeterministic; detector settings are free, and nothing in it permits faster-than-light signaling (§7.5).
- **The Born rule is not derived in this paper — and is now addressed in a companion.** The gap is stated once, in full, as the fork of §3.3 and carried as open problem 1 of §8; later mentions refer back to it rather than restate it. Since this paper’s review round, the companion selection paper [47] has derived the *fairness* of the outcome-selection game from stated detector premises (four theorems, conditional on a wave-realism premise and ordinary detector physics), leaving the square itself to Gleason’s theorem and to driven-oscillator energetics. This paper’s registration account neither consumes nor depends on that result; the §3.3 fork and §8 record what it changes.
- **Which outcome is selected is inherited, not derived.** We take the outcome as an input — exactly as standard conditional-measurement theory conditions on a record — and confine the framework’s contribution to what happens *after* selection: the reversible→irreversible registration (§3.1). A background-field configuration that would *determine* the outcome is offered only as a speculative candidate for the open selection law (§3.3, §8), never as the framework’s mechanism, and nothing in the registration account depends on it. We make this separation deliberately: the framework’s genuine content — the closed/open chiral dynamics and the registration reading — stands on its own and does not require solving the selection problem.
- **One candidate prediction, contingent on a postulate.** A linewidth-dependent gravitational Bell effect (§6). We show it does *not* follow from the framework plus standard QED (§6.2, Appendix C);

it requires an additional, non-covariant coupling. The experiment is a clean falsification test of that postulate, and is the framework's one observational handle.

- **The framework carries a preferred frame, and we say so.** In a single world, Bell nonlocality is anchored to a physical preferred frame — the bulk/vacuum rest frame — which enters only through measurement (Stage 2); the unitary dynamics stays exactly Lorentz covariant. Ordinary matter couples to that frame, beyond the electromagnetic locking, only gravitationally and so weakly. Demonstrating quantitatively that this evades the precision Lorentz tests is part of what the framework still owes (§8).
- **The chiral coupling $K = m$ is a fixed constant.** It is set once by the Higgs vacuum expectation value, a Lorentz scalar; gravity *reads* the chiral register (the mass term $\bar{\psi}\psi$ to which the stress tensor couples) but does not *modulate* K . There is no $K(\Phi)$ dependence, so the framework carries no equivalence-principle liability — only the preferred-frame (Lorentz) liability above.

2. Two regimes of the chiral mass coupling

2.1 Chiral decomposition

In the Weyl basis the Dirac spinor splits as $\psi = (\psi_L, \psi_R)^T$ and the Dirac equation [7] becomes the coupled pair

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m \psi_R, \quad i\sigma^\mu \partial_\mu \psi_R = m \psi_L,$$

with $\sigma^\mu = (I, \vec{\sigma})$, $\bar{\sigma}^\mu = (I, -\vec{\sigma})$. For $m = 0$ the two equations decouple and the particle has definite chirality; for $m \neq 0$ the mass couples the sectors. In the chiral basis the mass is the single off-diagonal entry of the Dirac Hamiltonian,

$$H = \begin{pmatrix} \vec{\sigma} \cdot \vec{k} & m \\ m & -\vec{\sigma} \cdot \vec{k} \end{pmatrix},$$

set by the Higgs–Yukawa coupling, $m = y_f \langle \varphi \rangle$ (in SI units the Compton frequency $mc^2/\hbar$). This identification is textbook; what follows is a reading of its dynamical consequences, and the central point is that the mass coupling is *not* itself a synchronization attractor.

2.2 The closed system: unitary precession, no attractor

Apply the polar (Madelung) decomposition $\psi_X = \sqrt{\rho_X} \cdot e^{i\varphi_X} \cdot \chi_X$ ($X = L, R$), with $\rho_X \geq 0$ a real amplitude density, φ_X a real phase, and χ_X a unit two-spinor carrying the spin orientation. In the rest frame ($\nabla \rightarrow 0$, χ constant), separating real and imaginary parts of the coupled equations gives

$$\dot{\phi}_L = -m\sqrt{\rho_R/\rho_L} \cos \Delta, \quad \dot{\rho}_L = +2m\sqrt{\rho_L\rho_R} \sin \Delta, \quad \Delta \equiv \phi_R - \phi_L,$$

and the mirror equations for R . Two features are decisive. First, the trigonometric functions land in the *opposite* places from the Kuramoto model [8]: the **sine sits in the amplitude equation and the cosine in the phase equation**, whereas Kuramoto carries its sine in the phase equation. Second, the phase-difference dynamics

$$\dot{\Delta} = -m \cos \Delta \frac{\rho_L - \rho_R}{\sqrt{\rho_L\rho_R}}$$

vanish on the symmetric manifold $\rho_L = \rho_R$ for *every* Δ . There is no restoring force and no preferred phase difference — a marginal line of fixed points, not a basin. The motion is normal-mode (Rabi) precession: defining $\psi_\pm = (\psi_L \pm \psi_R)/\sqrt{2}$ gives $i\partial_t \psi_\pm = \pm m \psi_\pm$, so the chiral populations flop coherently at beat frequency $2m$.

It is worth making explicit *why* the absence of a basin protects superposition, because the two statements are one fact in different language. Everything physical about a chiral superposition lives in the relative phase Δ (and the amplitude ratio); to *lock* is to have a restoring force that pins Δ to a preferred value Δ^* , and to pin Δ is precisely to destroy the superposition. The contrast with Adler/Kuramoto is exact: there $\dot{\Delta} = \Delta\omega - K \sin \Delta$ derives from a potential $V = -K \cos \Delta$ with a genuine *minimum* (a basin, a restoring force), whereas the cosine-in-phase / sine-in-amplitude pairing above is the signature of a *conservative* (Hamiltonian) flow, not a *gradient* (relaxational) one — it conserves a quantity rather than descending a potential. Geometrically, the mass term (a σ_x coupling) precesses the Bloch vector in closed circles *on the surface* of the sphere at the rate $2m$; locking would require it to spiral inward (toward a mixed state) or into a pole (a definite outcome), and unitary precession does neither, conserving purity exactly. (When the system is opened, that inward spiral acquires an explicit rate — $\Gamma/2 = -\text{Im } \Sigma$, the imaginary part of the dressed Dirac mass — derived in §3.6.)

This is not a missing feature; in the closed regime it *is* the physics. A closed, unitary system admits no Lyapunov function for an order parameter to climb — Liouville’s theorem makes the flow volume-preserving, so it has no attractors of any kind. The chiral phases can only precess, never settle. No relabeling rescues an attractor: redefining $\psi_R \rightarrow i\psi_R$ moves the sine into the phase equation, but its prefactor $[\sqrt{\rho_L/\rho_R} - \sqrt{\rho_R/\rho_L}]$ is odd under $L \leftrightarrow R$ and vanishes on $\rho_L = \rho_R$, so no single-signed $K \sin \Delta$ term survives. An attractor cannot arise from unitary evolution; a bath must be traced out.

We are careful about what this establishes. That an *isolated* system preserves superposition is, on its own, just unitarity — true of any closed system, and not something the chiral reduction is needed to prove; reading §2.2 as a *derivation* of superposition’s survival would be claiming credit for Liouville’s theorem. The content here is narrower and framework-specific. The mass term couples two phase sectors and superficially *resembles* a Kuramoto synchronizer, so one might fear it carries a built-in, always-on lock — in which case every massive particle would continuously self-measure even in perfect isolation, contradicting observed coherence and sinking the framework. The reduction shows it does not: the would-be synchronizer has an off-switch, and that switch is the closed/open boundary. This is therefore a *consistency (no-go) result* — the proposed collapse mechanism provably does **not** misfire in the one regime where collapse must not occur — rather than an explanation of why superposition survives.

Two scopes must stay distinct. The result above concerns the *chiral L/R relative phase of a single particle*. The *measurement* superposition (system $\otimes$ pointer) is protected by the same general principle — closed $\Rightarrow$ no attractor — but that protection follows from unitarity at the composite level, not from the chiral calculation specifically; we do not claim the single-particle reduction explains pointer coherence. Finally, the marginal line is not a dead end but the seam to §2.3: neutral stability means a small perturbation moves Δ freely along that line, neither resisted nor amplified — exactly the soft direction a bath couples to. Opening the system tilts the flat marginal line into a sloped potential with a minimum, the Adler basin of §2.3, so the off-switch and the on-switch are the same structure under two boundary conditions.

A numerical check in the two-body case. The reduction above concerns one particle’s chiral L/R phase, and the scope caveat notes that composite superpositions are protected by unitarity in general rather than by that calculation. The configuration for which the Kuramoto worry is most naturally posed falls between the two: *two* genuine oscillators with an explicit mutual interaction, which is the actual Kuramoto setting and which neither argument covers. That case can be tested directly. We evolved two Dirac particles in 1+1D, bound in a scalar harmonic well and coupled by a Breit $\alpha_1\alpha_2$ term, under closed unitary split-operator evolution, and asked whether their Zitterbewegung phases entrain.

They do not, and the failure has a mechanism. Turning on any coupling at all drops the mean reduced-state purity from 0.924 to ≈ 0.30 –0.40, where it plateaus; across twenty-five coupled, detuned parameter cells the phase-order parameter and the purity are anticorrelated at -0.77 , and the best simultaneous cell in the sweep reaches only $R = 0.444$ with purity 0.441. There is no accessible regime in which the two clocks are both coupled and individuated. The one cell that did show genuine drift suppression — the relative phase advancing at 0.39 of the bare detuning rate, which is what partial entrainment looks like — **loses it entirely when the evolution is lengthened fourfold**, returning to 1.10 of the detuning rate with the order parameter decaying from 0.839 to 0.518. What resembles an Arnold tongue is a transient correlation

that decays, not an attractor.

The mechanism is the one §2.2 predicts but does not name: **the coupling that would align the phases instead builds correlation between the subsystems**, and the phase information migrates out of the marginals and into the joint state. Closed-system chiral coupling has no attractor because its would-be locking strength is consumed generating entanglement. This is the same physics as thermal decoherence, which entangles a system with unobserved modes — so no separate thermal argument is needed, and none is available in a closed system anyway.

Method note, because the measurement is easy to get wrong. The oscillator phase must be read as a *state* variable — the Bloch azimuth of the reduced spin state — rather than inferred by band-pass filtering an observable and taking an analytic signal. The two agree to 0.008 in the uncoupled limit and diverge by up to 0.105 once coupled, with the filtered estimate **overstating** entrainment precisely in the cells where purity has collapsed and the phase is least well defined. Where the marginal is nearly maximally mixed the azimuth is not small but undefined, and any phase-based order parameter loses its domain there. The appropriate generalization for spinor-valued clocks is the non-Abelian Kuramoto model on $SU(2)$ [49], which reduces to the scalar case at $n = 1$.

Scope. Two bodies, one spatial dimension, one interaction, one family of initial conditions. This does not establish the closed-system no-go in general — that remains the analytic argument’s job — but it closes the specific gap where a mutual-coupling lock could have hidden, and it supplies the mechanism.

2.3 The open system: dissipation supplies the attractor

Couple the chiral pair (or a composite particle) to a bulk and trace the bulk out. The reduced phase obeys an Adler equation [9],

$$\dot{\phi} = \omega + K_{\text{eff}} \sin(\Phi_{\text{bulk}} - \phi),$$

which *does* possess a genuine attractor, because the bath now carries away the phase difference and a Lyapunov function exists.

The form above is the lock’s **first-harmonic limit**, with the bulk entering as a single biased reference phase Φ_{bulk} ; it is monostable — one locked phase, one outcome. The general dissipative coupling carries a second term. A bulk that *measures* the chiral interference channel — a symmetric, $\langle \cdot \rangle$ -free coupling even under $\phi \rightarrow \phi + \pi$ — contributes a second harmonic, so

$$\dot{\phi} = \omega - \gamma \sin 2\phi - \epsilon \sin(\phi - \Phi_{\text{bulk}}),$$

with γ the symmetric measurement rate and ϵ the polar bias of the line above. The anti-phase fixed point $\phi = \pi$ is stable iff $\epsilon < 2\gamma$: for a *faint* bias ($\epsilon \ll \gamma$ — the ppm thermal bias of §4.1) the lock is **bistable**, stabilizing phases near 0 and π (a binary), the bias merely deepening one well; for a *dominant* bias ($\epsilon > 2\gamma$) it collapses to the monostable first-harmonic lock. Both are limits of one equation. The two-outcome structure of a measurement is thus the symmetric (measurement) part of the bulk coupling; the bias selects among the basins but does not create them. Two limits on this claim must not be overdrawn, and Appendix D is the control on both. First, the bistability *stabilizes* a binary but fixes neither its cardinality (set by the measured observable’s spectrum) nor the Born weights (§8), both of which remain open. Second, a bistable phase flow supplies *structure*, not *registration*: a fixed point of the reduced phase equation is not an outcome — the same dissipator that locks the phase decays the transverse coherence carrying it, and an outcome is realized only along a conditional (monitored) trajectory in the einselected pointer basis, not in the phase variable alone (Appendix D). The antipodal basins are therefore the two-outcome *geometry* the chiral clock makes available (§3.8); the registered outcome lives in the conditioned pointer record.

This — not the closed mass term — is the framework’s synchronization step, and it places the picture within the established literature on quantum synchronization [10–13], where Kuramoto-form phase dynamics

are derived from the semiclassical limit of quantum master equations for genuinely self-sustained (limit-cycle, dissipative) oscillators. The two regimes are the same off-diagonal coupling under two boundary conditions: closed $\rightarrow$ precession (superposition preserved); open $\rightarrow$ Adler lock (measurement). Coupling and synchronization are therefore not rival mechanisms but the same coupling under two boundary conditions. The off-diagonal coupling is continuous across both regimes; the *attractor* is not — it is supplied by the bath. Where there is no bath the coupling can only precess; once a bath is present and traced out, the dissipation converts that precession into a locking attractor. The boundary between them — the onset of dissipative coupling to a bulk of overwhelming inertia — is the framework’s candidate for a physical Heisenberg cut, made precise as a condition on the effective action in §3.5. We use “synchronization” hereafter only in this strict dissipative sense.

2.4 Normal modes and the Zitterbewegung beat

Reinstating the kinetic term, diagonalizing H on a helicity block gives eigenvalues $\pm\sqrt{k^2 + m^2} = \pm E$: the relativistic dispersion. The normal-mode splitting is $2E$, equal to $2m$ at rest, and this is the Zitterbewegung beat.

We are explicit about a factor of two that is easily blurred. The Zitterbewegung beat is the splitting between the **positive- and negative-energy normal modes** ($+E$ and $-E$), which at rest is $2mc^2/\hbar$. It is *not* the difference of a “temporal clock” at $\omega_t = E/\hbar$ and a “spatial clock” at $\omega_s = p\cdot v/\hbar$; in the rest frame that difference tends to $mc^2/\hbar$, not $2mc^2/\hbar$, so the temporal/spatial pairing does not reproduce the beat. The two clocks whose interference gives Zitterbewegung are the $\pm E$ modes; the de Broglie carrier is their symmetric mode at $\omega = E/\hbar$. The Zitterbewegung beat is the empirical signature that the chiral coupling is real: without m , ψ_L and ψ_R free-stream and no beat forms.

Claim	Status
Mass couples $\psi_L \leftrightarrow \psi_R$; $m = 0$ decouples them	Standard QFT; textbook
Higgs–Yukawa sets the coupling, $m = y_f\langle\varphi\rangle$	Standard Model
Closed mass coupling = unitary precession, no attractor	This work (rigorous; §2.2)
Measurement = dissipative Adler/Kuramoto lock to a bulk	This work (interpretive; §3)

2.5 When the open regime engages: free flight, boundaries, and gradients

The two regimes raise a question §2.2–2.3 leave open: at any given moment, which one is acting? The answer is set not by time but by *place*. The open, dissipative regime — the only one we call synchronization — engages only where an external structure **deforms the wave’s amplitude**: a material boundary (an aperture edge, a detector surface), a field gradient (Stern–Gerlach, MRI), or spacetime curvature (a tidal gradient). In free flight through an unbiased, gradient-free vacuum, only the closed regime runs, and the particle stays coherent.

This is not a postulate but a consequence of what decoherence requires. A freely propagating *inertial* particle does not radiate; it deposits no real quantum in the vacuum, so there is no record to carry away which-path information and nothing to decohere it — which is why interferometers hold coherence over macroscopic paths. The one vacuum interaction always present in flight, the self-energy dressing that renormalizes the mass, is coherent and non-recording: it *maintains* $K = m$ rather than reading it. A record — and therefore a measurement — forms only where the particle accelerates or scatters, that is, at a boundary.

In flight, then, the dynamics is the closed-regime guidance already contained in the Madelung decomposition of §2.2, now read ontologically — the spatial gradient set aside in §2.2’s rest-frame reduction is restored here, and the phase guides through $\mathbf{v} = \nabla\varphi/m$, with the amplitude entering through the quantum-potential term (§7.3). We adopt this guidance law — it is the dynamical guiding law §7.3 records as owed — but locate the *outcome-selecting* randomness not in flight but in the unbiased background sampled **at the boundary** (§3.3); the $|\psi|^2$ form of that selection remains open (§8), with relaxation to quantum equilibrium [42, 43] the

candidate route. One caution against a hidden assumption: the in-flight guidance phase and the internal Zitterbewegung clock are not independent factors but the two normal modes of the $K = m$ coupling (§2.4) — the de Broglie carrier and the $\pm E$ beat. Treating their rates as decoupled is a rest-frame approximation, not a separate premise.

Boundaries come in two kinds, distinguished by whether a record is taken. A **coherent** boundary — an aperture, or a Stern–Gerlach gradient — reshapes or splits the wave while taking no which-path record; it stays in the closed regime, which is exactly why two-slit interference survives the slits and a Stern–Gerlach beam can be recombined. A **recording** boundary — a detector surface, an absorbing screen, a readout coil — exchanges real quanta and fires the open regime. A field gradient (Stern–Gerlach, MRI) is a *biased* boundary: an applied field, not the unbiased vacuum, sets the reference axis, yet the irreversible record still forms only at the screen or coil. Gravity sits at the far weak end: a uniform potential is removable by the equivalence principle and does nothing in flight, and only a curvature/tidal gradient is a genuine in-flight gravitational deformation — equivalence-principle-safe, since it does not modulate the fixed K (§1.3), but negligible in the laboratory and relevant only in strong-field astrophysics.

3. Measurement as resonant capture and bulk re-synchronization

3.1 Three stages: capture, selection, registration

A measurement separates into three steps, consistent with the locality of fundamental interactions. (Earlier versions of this paper, and the frozen demonstration snapshot, used a two-stage division; the present form splits the old Stage 2 at a boundary the formalism already carried — the two clauses of §3.5’s joint condition, and the amplification threshold of Appendix D. Where companion documents say “Stage 2,” read Stages 2–3 below.)

- **Stage 1 — resonant capture (reversible).** The incoming system phase-couples to a single bulk-bound partner at the interaction vertex (a photon to a bound electron, a spin to a detector dipole). The vertex is electromagnetic — a charge-current coupling, with energy entering as a *selection rule* rather than as a coupled density (§3.7): the capture rate is the cross-section times the local intensity, $r \propto \sigma(\omega) |\psi|^2$, so the resonance filter $\sigma(\omega)$ makes the step energy-selective while the drive is amplitude- (hence interference-) sensitive. This is the regime of injection locking and of the laser: constructive (resonant) interaction selects which mode the system couples to, governed by the Arnold-tongue condition $|\Delta\omega| \lesssim K_{\text{pair}}$ — mismatches large compared to the coupling are not pulled in. Capture shapes which outcome channels participate — the measured basis — and records nothing; while only this stage operates (Re W alone, §3.5), the evolution is reversible and any dephasing refocusable.
- **Stage 2 — selection (the surface commit).** The capture partner is *never a closed system*, and its dissipation begins with an interaction that accounts of measurement routinely pass over: the excited electron’s continuing coupling to its own parent atom. A photoelectron is not born free. It leaves an ion whose Coulomb field it still feels; the ion recoils and takes up momentum; the atom’s remaining electrons rearrange (shake-up, Auger); and atom and electron sit in a lattice that absorbs the difference as phonons. That excited-electron–host-atom interaction is the first dissipative channel — the door through which the local environment’s fluctuations act on the near-threshold dynamics — and it is here, still at the surface, that the run is committed: the one actual background configuration (§3.3, §4) tips the flow into a single basin. Formally this is the onset of the absorptive part ($\text{Im } W \neq 0$, §3.5): a real, on-shell excitation now exists. But selection-in-progress is not yet a fact. Within this window the dynamics remains reversible in principle — the catch-and-reverse regime [14], the refocusable side of §3.2, the decay of sub-threshold latent-image specks (§5). The Born weights are *set* by the Stage-2 dynamics; they become *final* only at Stage 3.
- **Stage 3 — registration (reservoir-powered closure).** The committed excitation is closed out by dissipation into a continuum of environmental modes from which it does not recur. In an ordinary, non-recording interaction this is plain **relaxation toward local equilibrium** — fluorescence, phonon emission, the electron settling back toward the bulk average exactly as a spin relaxes toward B_0 in NMR (T1/T2). In a detector built to *record*, it is **amplification**: the event triggers a pre-loaded free-

energy reservoir — the dynode chain’s applied voltage, a developer bath’s chemical potential, a cloud chamber’s supersaturation — which blows the single excitation up into a macroscopic, readable record. Both are dissipative and one-way; amplification only renders the already-committed event permanent and readable. The registration step is *charge-triggered and energy-blind*: a photomultiplier delivers the same 10^6 -electron pulse whether the photon that started the cascade carried 2 eV or 3 eV, because the amplifier couples to the presence of liberated charge (j^0 , §3.7), not to the energy that liberated it. Registration is therefore equivariant by design — it copies whichever outcome Stage 2 committed and contributes no statistics of its own. Its irreversibility is *erasure*: the phase mismatch carrying the system’s prior coherence is scattered into the bulk’s uncontrolled modes beyond practical recall, so the system cannot recohere and the record is stable. The erasure is *effective*, not fundamental — consistent with the no-collapse commitment the global evolution remains unitary and the history is scrambled into the bulk rather than destroyed; what is gone is the system’s *locally recoverable* history, the irreversible T1/T2 side of §3.2 and not the refocusable T2* side. This is the framework’s through-line, and it is an *information* statement: the wavefunction does not collapse but disperses — the incident state’s information spreads across the reservoir’s many particles, so the *total* fine-grained entropy is conserved while the *locally accessible* entropy rises. What looks like collapse is that dispersal: the outcome value copied redundantly (making the record objective, the quantum-Darwinism face) and the relative phase scrambled beyond recall (making it irreversible, the scrambling face) — never a physical destruction of ψ .

Three things are therefore distinct and should not be run together, and the three stages give each its own address: the *selection* of a single definite outcome (Stage 2 — the role of the background configuration, §3.3, not of the dissipation itself); the *decoherence* the dissipative coupling produces along the way (a decohered, mixed reduced state); and the *erasure* that stabilizes the record (Stage 3). That an *irreversible act of amplification* is what closes a measurement is an old observation — it is Bohr’s irreversibility requirement [21] and the Bohr–Wheeler dictum that “no elementary phenomenon is a phenomenon until it is a registered phenomenon, brought to a close by an irreversible act of amplification” [22] — and the macroscopic-equilibration mechanism we invoke for Stage 3 is closely modeled on the ergodic apparatus-relaxation account of Daneri, Loinger, and Prosperi [23]. Our contribution is not the irreversibility, which is standard, but its identification with a dissipative phase-locking (Adler) step and the separation of the *selecting* (Stage 2) from the *registering* (Stage 3) role — a separation Bohr’s dictum runs together.

All three stages are dissipative in different degrees, separated by hierarchies of *timescale and threshold* rather than by a sharp unitary/dissipative ontological boundary: the capture partner is never truly closed, so Stage 1 carries a weak dissipative component from the outset; Stage 2 is the absorptive onset; Stage 3 is the point of no return. The honest statement remains a separation of rates, not of kinds — now with the two boundaries between the stages identified with the two clauses of §3.5’s joint condition ($\text{Im } W \neq 0$; $\Gamma_{\text{cap}} > 0$, above the amplification threshold of Appendix D).

3.2 The reversible→irreversible boundary is demonstrable

The content of “fast capture, slow equilibration” is that there is a window in which the early dynamics are reversible and a point beyond which they are not. This is not a metaphor; three independent systems exhibit it directly:

- **The nuclear-spin echo.** In NMR/MRI the transverse dephasing that follows a radiofrequency pulse separates into a *reversible* part (T2, *from static field inhomogeneity, refocused by a 180° echo*) and an irreversible* part (T2/T1, stochastic spin–spin and spin–lattice relaxation). The spin echo is a routine, unambiguous reversal of the capture-stage dephasing — a Humpty-Dumpty experiment for spins.
- **Catch-and-reverse of a qubit.** Mineev et al. [14] track a quantum jump in a superconducting transmon and reverse it mid-flight with ~82% fidelity, in a regime where the dissipative readout channel has been engineered to near-zero. In framework terms: with the Stage-3 (readout) coupling suppressed, no record forms and the Stage-2 window stays open — evolution reversible until the first real dissipative event.
- **Stern–Gerlach recombination.** Carefully recombined SG paths recover the original superposition: the gradient creates the correlation, not the outcome; the outcome arrives only at the irreversible

detection.

3.3 Why outcomes are definite — and what this does and does not explain

The detector’s macroscopic inertia provides the asymmetry that makes the record definite. When a small oscillator (mass m) couples dissipatively to a large one ($M \gg m$), the small one conforms; the reverse is suppressed by m/M . This is what the framework actually delivers: *given* a selected outcome, the dissipative flow carries the state into exactly one attracting basin and holds it there — one ψ , one record, one world, with no projection postulate and no observer. *Which* basin is the selected one we **inherit**, exactly as the standard conditional-trajectory unraveling (Appendix D) conditions on the meter’s vacuum noise dW ; we do not derive it. A stronger reading — that a single actual configuration of the background field (§4) *determines* which basin, completing that noise with a hidden variable — is the framework’s **speculative candidate** for the open selection law (open problem 2 of §8), offered and not asserted. The two readings are not yet reconciled (Appendix D, point 2); the registration account rests only on the inherited one.

We are deliberate about what this does *not* settle. The attractor explains why there is exactly **one** outcome per run; it does not explain why the long-run **frequency** of each basin is $|\alpha|^2$. That is the Born measure, stated in full here and carried as open problem 1 of §8. Conflating the two — treating “definite outcome” and “correct weight” as one result — is the error we most want to avoid.

Within that speculative candidate, the *oneness* half of definiteness — exactly one outcome across an extended detector, however many sites the wave reaches — carries its own candidate sub-mechanism rather than a stipulation: **conservation-enforced winner-take-all**. (This, with its nonlocal depletion below, belongs to the candidate selection law, not to the inherited-selection registration account; we present it as a proposal to be judged, and the framework’s core does not depend on it. *v8*: the proposal has since been developed into the core engine of the companion selection paper [47] — energy-budget depletion with first-closure statistics, where energy conservation makes registration rates kinematically linear in occupation (its Theorems 4–5), rendering the statistics exact at every registration speed and in every quantum sector. It is therefore no longer only a proposal; its status there is proved-conditional, on stated premises including, for the entangled sector, an explicit shared-registry ontology.) The packet carries one quantum; a site’s threshold requires the whole quantum; a completed absorption therefore *is* the depletion of the budget, and no second site can close. Three qualifications keep this honest. (i) The depletion is **nonlocal**: the packet’s remote amplitude — which can be half the norm in a spacelike-separated interferometer arm — must be drained on the preferred-frame foliation (§8). The mechanism thus *physicalizes* the collapse as a real, foliation-synchronized energy transfer rather than eliminating it; single-photon anticorrelation persists at spacelike detector separation [45], so the enforcement is superluminal, and the framework’s existing nonlocality (§7.5) is restated, not removed — with one structural gain: the preferred frame, elsewhere carried as a cost, here does necessary work. (ii) The local step is a **rate, not an accumulation**: the commit probability per site goes as the locally delivered power (§3.1), while the budget decides how many commits can close; a local energy-accumulation reading would predict intensity-dependent first-click latency, excluded since the earliest fast-timing photoemission measurements [46]. (iii) The drain is **provisional at Stage 2 and final at Stage 3**: a committed excitation can still re-emit (resonance fluorescence; radiation trapping in dense media) and return the loan — absorption is not yet detection — and a completed drain leaves the field mode in the vacuum state *exactly*, with no sub-threshold real residue (any energy mismatch departs as real, local quanta at the winning site — photoelectron kinetic energy, phonons — never as a lingering field remainder; the first version’s residual-thermalization claim stays retracted). The recorded statistics are therefore first-*closure*, not first-touch, statistics — Born-weighted provided registration is equivariant (§3.1). For the ledger of §8’s substrate-sampling model, this is an upgrade: winner-take-all is no longer imported by fiat; its residue is the foliation-synchronized budget — the nonlocality already owned, given a physical carrier.

Stated once, as the fork it forces — the canonical form to which every later mention of the “Born gap” refers back: *either* the basin weights are **derived** from the synchronization dynamics — in which case the minimal two-outcome model settles it, and its locking basins have area 50/50, flat in $|\alpha|^2$ (the attractor geometry carries no amplitude information about the prepared state) — *or* $|\alpha|^2$ is **imported** as the measure on the per-run background, a quantum-equilibrium postulate of exactly Bohm’s status. The framework takes the **second horn**, and we state it plainly: the dissipative lock supplies *registration* (one definite outcome, §3.1–

3.5) and, where the Stage-2/3 coupling is non-demolition, *equivariance* (a measure granted at preparation is carried intact to the outcome, §3.7); it does **not** supply the weights, which enter as the measure placed on the selecting ensemble (§7.3, §8). MCI is therefore Born-*compatible*, never Born-*derivative* — the same status the other ψ -ontic single-world readings occupy, and an open problem we neither hide nor claim to solve.

Status update (v8). The fork as stated offered two horns — weights derived from the synchronization dynamics (wrong: the basin geometry is flat), or imported as a quantum-equilibrium measure (Bohm’s status). The companion selection paper [47] identifies a third route the fork did not anticipate, and takes it: the weight is carried neither by basin geometry nor by an imported measure but by the **deposited energy**. An incident quantum excites every candidate absorber site in proportion to squared local amplitude — driven-oscillator energetics, the only place the square enters — and a noise-driven exchange then selects the registering site as a *proved fair game* (a martingale closed by optional stopping), making registration frequencies equal initial energy shares, i.e. Born weights. The fairness is derived from detector premises rather than engineered into the noise; the result is conditional on those premises (wave realism; amplitude-linear coupling; the sub-threshold slaved-phase theorem with its κ_{ret} ansatz; noise locality; energy conservation). Within this paper, standing alone, the second horn remains the honest statement; the series as a whole now carries [47]’s conditional discharge of it, and open problem 1 of §8 is restated accordingly.

3.4 Decoherence as synchronization to the bulk

In the standard account a measured system loses *quantum* coherence with its entangled partner — the off-diagonal terms of its reduced density matrix decay. The synchronization picture names what replaces it: the system *frequency-locks* (entrains) to the bulk reference, acquiring a definite *classical* phase relationship to Φ_{bulk} . These are two different quantities, not one conserved quantity — the quantum coherence is genuinely destroyed *in the reduced (local) state*, while classical phase-locking is established — so the transition is the loss of one and the onset of another, not a transfer. (Locally genuine, globally unitary: the local coherence is gone but the global evolution is the no-collapse, history-into- the-bulk scrambling of §3.1.) Because a single run carries one actual background configuration (§3.3), the result is a definite state rather than an improper mixture that merely looks classical. This supplies einselection [4] with a specific dynamical mechanism — phase-locking stability — for why pointer states survive: they are the states that can entrain to the bulk.

3.5 The reversible→irreversible boundary, formally: $\text{Re } W$ and $\text{Im } W$

The three stages of §3.1 and the demonstrations of §3.2 have a compact formal counterpart. Integrating out the Dirac field in the presence of a background — an external field together with the conductor’s boundary conditions — yields the vacuum persistence amplitude

$$\langle 0_{\text{out}} | 0_{\text{in}} \rangle = e^{iW}, \quad |\langle 0_{\text{out}} | 0_{\text{in}} \rangle|^2 = e^{-2\text{Im } W},$$

with W the one-loop effective action. The probability that no real quantum is produced — that the in-state is *returned* — is $e^{-2\text{Im } W}$; the probability that a real, on-shell quantum appears is $1 - e^{-2\text{Im } W} \approx 2\text{Im } W$ in the weak regime. The split of W into real and imaginary parts is the reversible→irreversible boundary:

- $\text{Re } W$ — **Stage 1, reversible.** The real, dispersive part carries the virtual-pair physics: vacuum polarization and charge screening, the image-charge and Casimir energetics near the conductor, and the resulting level shifts. These are *reactive* — they renormalize energies and dress the state, produce no on-shell quanta, and leave no record; they are even under time reversal. This is the catch-and-reverse window of §3.2: a fluctuation that borrows energy ΔE and repays it within $\hbar/\Delta E$. While only $\text{Re } W$ operates, the evolution is unitary and any dephasing is refocusable.
- $\text{Im } W$ — **Stages 2–3, absorptive.** The imaginary, absorptive part is nonzero *only* when a channel can go on-shell, and it *is* the rate at which real quanta appear — the decay rate of the in-state; it is odd under time reversal. For a constant field it is the Schwinger result,

$$\frac{2 \operatorname{Im} W}{V_4} = \frac{e^2 E^2}{4\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^2} \exp\left(-\frac{n\pi E_{\text{crit}}}{E}\right), \quad E_{\text{crit}} = \frac{m^2 c^3}{e\hbar},$$

exponentially suppressed below the critical field. “Mostly reversible (Re), rarely real (Im)” is therefore exact and quantitative: the rare event is the exponentially small $\operatorname{Im} W$, and its threshold scale E_{crit} is set by the mass.

Producing a real pair is necessary but not sufficient for a record; the record is *completed* when the on-shell quantum couples dissipatively to the bulk — the Adler lock of §2.3, which in the three-stage division of §3.1 is precisely the Stage 2 $\rightarrow$ Stage 3 passage. The electron relaxes into the conductor’s electron sea at a capture rate $\Gamma_{\text{cap}} \sim K_{\text{eff}}$, the positron annihilates at $\Gamma_{\text{ann}} \propto n_e |\psi(0)|^2$ (overlap with the electron density), and the available final states are restricted by Pauli blocking through the Fermi factors $(1-f)$. A dense bulk does not *reverse* a real pair — a frequent misreading — it supplies the partners and dissipation that *complete* its irreversibility quickly.

The same Re/Im structure governs the general measurement, not only pair creation. For a system coupled to a detector bath the analogous object is the Feynman–Vernon influence functional $e^{i\Phi}$: $\operatorname{Re} \Phi$ renormalizes and shifts the system dynamics (reversible), while $\operatorname{Im} \Phi$ is the decoherence functional. For the linear, Gaussian coupling of §4.3 the latter is exactly the variance $\langle \delta^2 \rangle / 2$, so the visibility envelope $e^{-\langle \delta^2 \rangle / 2}$ is $e^{-\operatorname{Im} \Phi}$. The thermal source of §4.3 is a standard contribution to this $\operatorname{Im} \Phi$; the gravitational source of §6 would be a *second* contribution to the same envelope **only under the added postulate of §6.2**, and is absent without it — under standard QED the redshift cancels as a common-mode phase (Appendix C). (The vacuum persistence amplitude and the influence functional are distinct objects; we invoke them as two instances of one structure — reversible = Re, irreversible = Im — not as the same functional.)

The Heisenberg cut, made precise. The cut promised in §2.3 is the boundary between the regime in which only $\operatorname{Re} W$ (equivalently $\operatorname{Re} \Phi$) acts and the regime in which an imaginary part couples to an irreversible bath. It is *not* a fixed length, mass, or location; it is the joint condition

$$\operatorname{Im} W \text{ (or } \operatorname{Im} \Phi) \neq 0 \quad \text{and} \quad \Gamma_{\text{cap}} > 0,$$

an on-shell channel open *and* the resulting quantum dissipatively coupled to the bulk. The conjunction resolves the three stages of §3.1: $\operatorname{Re} W$ only — Stage 1; $\operatorname{Im} W \neq 0$ with the bath decoupled — Stage 2, the catch-and-reverse window; both clauses — Stage 3, the record. Either condition alone leaves the evolution reversible: an on-shell quantum with the bath removed ($\Gamma_{\text{cap}} \rightarrow 0$) is the engineered catch-and-reverse regime [14] — Stage 2 without Stage 3 — where $\operatorname{Im} W$ may be nonzero yet no record forms; and a state dressed only by $\operatorname{Re} W$ never leaves the reversible window. Isolating or cooling the system *recedes* the cut; bringing a dense bulk to the interaction vertex *snaps* it there. The cut is thus the dynamical onset of absorptive coupling — exactly the closed $\rightarrow$ open transition of §2. A companion paper [48] develops this into a quantitative claim: the cut located where the sub-threshold phase-restoring rate falls to the coupling, $\kappa_{\text{ret}} \sim K$, with a computable fractional width $w = K/\omega$ spanning $\sim 10^{-8}$ (atomic absorbers) to $\sim 10^{-2}$ (broadband solid-state) — and derives the conventional cut’s celebrated movability as a theorem of the reversible sector rather than an embarrassment.

We are explicit about provenance. $\operatorname{Re} W$, $\operatorname{Im} W$, the Schwinger rate, the influence functional, and the Lindblad capture are standard results of quantum electrodynamics and open-system theory; the framework adopts them rather than deriving them. Its content here is the *identification* — Stage 1 = $\operatorname{Re} W$, Stage 2 = the onset of $\operatorname{Im} W$, Stage 3 = $\operatorname{Im} W$ gated by Γ_{cap} , the Heisenberg cut as the onset of absorptive bath coupling — which turns the reversible $\rightarrow$ irreversible boundary of §3.2 from a set of demonstrations into a condition with an explicit order parameter: the record probability $2 \operatorname{Im} W$, gated by the bath lock Γ_{cap} .

Read this way, the detector is a *condensed-matter* system and the pointer is a collective degree of freedom of it: the record probability $2 \operatorname{Im} W$ is its order parameter, and the Heisenberg cut is the driven–dissipative locking transition at which that order parameter becomes nonzero — a transition of the same family as quantum synchronization [10–13] and dissipative phase transitions more broadly, not a new postulate. The

collectivity resides in the detector's bath coupling Γ_{cap} , consistent with the einselection of pointer states [4, 5] that §3.4 supplies dynamically.

3.6 The single-fermion cut: the dressed Dirac mass

The collective order parameter of §3.5 admits a one-particle *reduction* — the many-body content sits in the bath coupling Γ_{cap} that is traced out below, not in ψ — and writing it in that form collapses §2.2, §2.3, and §3.5 into a single statement about where the Dirac mass pole sits. Couple the fermion to the bulk and trace the bath out; the propagator acquires a self-energy Σ , and the pole — the physical mass — moves:

$$S(p) = \frac{i}{\not{p} - m - \Sigma(p)}, \quad \Sigma(p) = \text{Re } \Sigma - \frac{i}{2} \Gamma, \quad m_{\text{eff}} = m + \text{Re } \Sigma - \frac{i}{2} \Gamma.$$

That single complex number carries the reversible and absorptive sides of the three stages:

- $\text{Re } \Sigma$ — **Stage 1, reversible.** A shift of the mass, hence of the zitterbewegung beat frequency $2m_{\text{eff}}c^2/\hbar$. The fermion is *dressed* — its clock redialed — but undamped: the §2.2 regime of a real mass, with the chiral $L \leftrightarrow R$ Bloch vector precessing on the surface of the sphere at conserved purity.
- $\text{Im } \Sigma = -\Gamma/2$ — **Stages 2–3, absorptive.** The pole leaves the real axis. In the chiral/Bloch picture $m \rightarrow m - \frac{i}{2}\Gamma$ is non-Hermitian: the precession no longer circles the sphere but **spirals inward at rate $\Gamma/2$** — exactly the inward spiral §2.2 identified as the signature of locking, now supplied by the bath. $\text{Im } \Sigma$ is the strength of the Adler attractor of §2.3.

Two standard results pin Γ , and they meet the framework's two ingredients. The *rate* is the golden rule, $\Gamma(\omega) = 2\pi g_{\text{eff}}^2 J(\omega)$: the decay rate is set by the bath spectral density J at the system frequency, so $\text{Im } \Sigma \neq 0$ only if the bath carries real modes resonant with the fermion — the on-shell condition of §3.5 as a condition on J . The *gate* is the Arnold tongue: linearizing the Adler equation $\dot{\phi} = \omega + K_{\text{eff}} \sin(\Phi_{\text{bulk}} - \phi)$ about the locked phase Δ_* gives $\delta\dot{\phi} = -K_{\text{eff}} \cos \Delta_* \delta\phi$, so the lock relaxation rate is

$$\Gamma = K_{\text{eff}} \cos \Delta_* \xrightarrow{\text{resonance}} K_{\text{eff}}, \quad \text{a lock existing only inside } |\Delta\omega| \lesssim K_{\text{eff}}.$$

Hence $\text{Im } \Sigma = -\frac{1}{2}K_{\text{eff}} \cos \Delta_*$: the bath coupling *is* the imaginary part of the dressed mass, and the Arnold-tongue capture condition of §3.1 and the golden-rule “ $J(\omega) \neq 0$ ” are the same gate.

Everything then reduces to one inequality on a pole — the single-fermion form of the §3.5 joint condition:

$$\text{cut crossed} \iff \Gamma = -2\text{Im } \Sigma > 0 \iff \underbrace{|\Delta\omega| \lesssim K_{\text{eff}}}_{\text{on-shell / Arnold tongue}} \quad \wedge \quad \underbrace{K_{\text{eff}} > 0}_{\text{coupled}}.$$

Closed or off-resonance, the pole stays on the real axis: real m_{eff} , reversible chiral precession (Stage 1). Opened and resonant, the pole moves off-axis: complex m_{eff} , the precession spirals into the lock (Stages 2–3). Because Σ is causal, $\text{Re } \Sigma$ and $\text{Im } \Sigma$ are Kramers–Kronig partners — the reversible redialing of the clock and its irreversible damping are two faces of one analytic object. The **Heisenberg cut is the dressed Dirac mass pole leaving the real axis.**

We are explicit about provenance and scope, as in §3.5. The self-energy and pole structure, the golden rule, Kramers–Kronig, and the Adler linearization are standard; the framework's content is the *identification* $\text{Im } \Sigma = -\frac{1}{2}K_{\text{eff}} \cos \Delta_*$ — that the bath coupling entering as $\text{Im } \Sigma$ is the Adler lock rate. Appendix D works this through in a standard dispersive-readout (circuit-QED) model: it computes $K_{\text{eff}} = 4\chi^2\bar{n}/\kappa$ from the dispersive shift χ , readout occupation $\bar{n}$, and linewidth κ , and — via an explicit conditional (monitored) trajectory — shows that this rate sets how fast a *single run* localizes to a definite pointer, while the *unconditional* dynamics yields only decoherence, not an outcome. The lock-rate identification is thus exhibited concretely; the outcome is supplied by the measurement record, not by the locked phase (the correction this revision makes to the appendix's earlier reading). We also keep Σ to its dominant scalar (mass-dressing) channel; the full self-energy is a 4×4 matrix with additional Lorentz structure, and we do

not claim that general decomposition here; §3.7 makes that restriction precise, and shows it runs the wrong way for the electromagnetic coupling.

3.7 What the measurement coupling can and cannot supply

The equivariance step that §8 will ask of Stages 2–3 — that the lock carry the prepared $|\cdot|^2$ weight intact to the outcome — presupposes that the Stage-2/3 coupling is *non-demolition* for the measured observable: the dissipator must preserve the populations of the measured projector while scrambling its phase. It is worth stating which observable the framework’s own coupling actually treats this way, because the answer is not the one the chiral picture invites.

In the chiral reduction the interference channel is the scalar bilinear $\bar{\psi}\psi = \sigma_x$ — the mass channel — whose two pointer phases $0, \pi$ are the binary of §3.2. A non-demolition measurement of it requires the bath to couple *through* σ_x . But the electromagnetic interaction couples through the **vector current** $\bar{\psi}\gamma^\mu\psi$, and the chiral classification of Dirac bilinears is exact: vector and axial currents are *chirality-preserving* ($\bar{\psi}_L\Gamma\psi_L$), while the scalar, pseudoscalar, and tensor bilinears are *chirality-flipping* ($\bar{\psi}_L\Gamma\psi_R$). The gauge field therefore couples only to the charge density ($j^0 = 1$) and the chirality-diagonal current (σ_z); it cannot generate the $L \leftrightarrow R$ mixing $\bar{\psi}\psi$. (The magnetic-moment, or Pauli, term is chirality-flipping, but it acts on *spin* — coupling spin to $\mathbf{B}$, as in Stern–Gerlach — not on the chiral scalar.) Equivalently at the self-energy level: in $\Sigma = \Sigma_V(p^2)\not{p} + \Sigma_S(p^2)m$ the scalar part $\Sigma_S \propto m$ is protected by chiral symmetry ($\Sigma_S \rightarrow 0$ as $m \rightarrow 0$), so the absorptive piece $\text{Im } \Sigma_S$ — the mass-channel attractor invoked in §3.6 — is the *chirally suppressed* one, while the unsuppressed dissipation lives in the vector channel $\text{Im } \Sigma_V$.

The consequence is concrete. Carried through to a basin split, the σ_z coupling reproduces no Born weights: for the rest-frame mass Hamiltonian it is transverse and drives *relaxation* toward the σ_x ground state (one pointer at $T = 0$, a Boltzmann mixture at $T > 0$); away from rest it is longitudinal and *dephases* in the chirality basis. In every case the outcome probability becomes independent of the prepared phase — the input weight is erased, the opposite of equivariance. The electromagnetic coupling is thus non-demolition for the charge/position/chirality pointer basis (ordinary which-path einselection, §3.4, which we keep) but *not* for the chiral interference channel. The two-basin σ_x structure is, on this analysis, a feature of the Stage-1 internal clock — the zitterbewegung beat of §2.4 — rather than the pointer basis the environment selects. We therefore do not treat the chiral $0/\pi$ binary as the universal measurement register. In ordinary detectors the environment selects the **charge/position (which-path) basis** — the basis the electromagnetic coupling *is* non-demolition for, and the one a cloud-chamber track or a calorimeter actually records; the chiral interference σ_x is internal-clock dynamics the environment does not monitor. The three-stage account of selection, registration, and irreversibility (§3.1–3.6) therefore rests on the which-path basis and does **not** require the chiral binary to be the pointer — the binary is the framework’s internal-clock structure, not its claim about what detectors record.

The single coupling that *is* non-demolition for $\bar{\psi}\psi$ is one that couples to the scalar/mass density directly — a Yukawa scalar, or gravity, since $\bar{\psi}\psi$ is (on the equations of motion) the trace of the stress–energy tensor, $T^\mu{}_\mu = m\bar{\psi}\psi$. This is the rigorous sense in which the Born-compatible coupling is gravitational, and it carries two costs that §7.4 and §8 take up: the Penrose self-energy timescale $\hbar/E_G$ is negligible for microscopic systems ($\sim 10^{16}$ – 10^{24} s for an atom or electron), becoming fast only at macroscopic pointer masses; and the objective-collapse programs that use this same operator do not *derive* the Born weights either, but normalize them in. We therefore do not claim that the framework’s coupling supplies the measured channel equivariance needs: the electromagnetic interaction supplies a different one, and the coupling that would supply it is gravitational, rate-suppressed, and no better off than Penrose–Diósi on the weights themselves. The Born measure (§8) is, on this analysis, uniformly open across the synchronization and gravitational-reduction readings alike.

One consequence of the charge-versus-mass split deserves statement as a result rather than an aside, because it converts a limitation into an explanation. **Every Standard-Model detection channel couples to a conserved current proportional to $|\psi|^2$; only gravity reads the energy-weighted density $T^{00} = E|\psi|^2$.** A charge-coupled detector therefore reports Born weights $|\psi|^2$ *independently of the channel energy* — verified numerically in `code/detector_resonance_selection.py`, where a charge readout of an unequal-

energy superposition returns the Born distribution while an energy-coupled readout returns $E_n|c_n|^2$. The equal-energy restriction on the wave-energy reading of the Born rule (§7.3, §8) is thus a *consequence* of measurement being electromagnetic, not a separate limitation awaiting removal — and it explains structurally why the framework’s one ontological discriminator (§6.4) is gravitational: gravity is the only interaction for which the real-field and probability-amplitude readings differ.

3.8 What the chiral coupling still does — and what it does not

Taken together, §3.7 and Appendix D force a conclusion better stated than left implicit: **the chiral mass coupling is not the engine of measurement in real detectors**. The Stage-2/3 mechanics — the dissipative Adler lock, the conditional selection, the amplification threshold — is generic open-system physics: any two-level pointer dispersively coupled to a damped mode exhibits it (Appendix D is, deliberately, a standard circuit-QED model), and the basis it runs in is the charge/which-path basis the electromagnetic coupling einselects (§3.7), which requires no Dirac structure at all. What, then, does $K = m$ still buy? Four things, in descending order of load borne:

1. **The two-regime no-go (§2.2).** The demonstration that the mass term — the one coupling that superficially resembles a built-in synchronizer — provably cannot lock in a closed system is a statement *about the chiral coupling itself*, and it is what licenses the framework’s central claim that locking is exclusively an open-system, boundary phenomenon (§2.5).
2. **The physical clock (§2.4).** The internal phase whose entrainment the interpretation is named for is not postulated: it is the $\pm E$ normal-mode beat of the chiral coupling — the Zitterbewegung — with its rate set by $K = m$. The ontology (a real field carrying a real clock) comes from the Dirac structure even where the detector dynamics does not.
3. **The binary geometry (§2.3).** The second-harmonic (apolar) structure of the chiral interference channel supplies a natural antipodal two-outcome geometry — available to a coupling that reads the scalar density $\bar{\psi}\psi$, i.e. a Yukawa or gravitational measurement (§3.7), but not exercised by electromagnetic detectors.
4. **The dressed-mass form of the cut (§3.6).** The identification of the Heisenberg cut with the Dirac mass pole leaving the real axis is stated in chiral variables; its content — $\text{Im } \Sigma$ as the Adler lock rate — transfers to any pointer, the chiral form being the fermionic instance.

The two halves of this paper’s subtitle therefore stand in an asymmetric relation, and we state it plainly. The *two regimes* are regimes of the chiral coupling — that analysis (§2) is the framework’s own. The *measurement mechanism* runs, in every laboratory detector, on the electromagnetically-einselected which-path basis (§3.7), where the chiral register is a passenger. The chiral coupling supplies the framework’s ontology, its clock, and its no-go discipline — not its pointer.

4. The bulk reference

4.1 What the bulk is: a faint thermal bias, not a coherent macrostate

It is essential not to overstate the bulk’s coherence. The reference is **not** a macroscopically entangled or coherent quantum state. The cleanest quantitative handle is magnetic resonance: the equilibrium nuclear polarization that defines the bulk reference axis in MRI is $P \approx \hbar\gamma B_0/2k_{\text{BT}} \approx 5$ parts per million at 1.5 T and body temperature. The bulk is ~99.999% thermal, disordered, and mixed; the “reference” is a faint statistical bias on top of a hot reservoir. That faint bias is nonetheless sufficient to image the proton distribution of the body — existence proof that a measurement reference need not be a coherent macrostate.

We therefore drop the language of an “entangled bulk.” The reference is a *phase-synchronized bias*, maintained by whatever interaction dominates locally.

Two attributes of that bias must be held apart, because temperature treats them differently. Write the reference as a complex order parameter $r e^{i\psi}$: a *magnitude* r — how large the bias is, the parts-per-million above — and a *phase* ψ — the reference *axis*, the direction the bias points, fixed by the dominant interaction

(the direction of B_0 in MRI). Temperature suppresses the magnitude but not the axis. The ~99.999% thermal disorder is exactly what holds r down to ppm; it does **not** erase ψ , which is pinned by the deterministic locking. The bulk is therefore *faint but oriented* — small amplitude, definite axis — and these are independent properties, a point §4.3 makes dynamical. That the axis ψ survives what temperature destroys is not incidental: the axis is the coherent content any reference must carry — a vanishing amplitude r can still serve, but a bath with no definite axis is no reference at all (§4.2).

4.2 Absolute phase is gauge; only differences are physical

A natural objection is that the bulk’s phase Φ_{bulk} cannot be measured — any probe prepared with a known phase synchronizes to the bulk on contact, so its absolute value is hidden by construction. This is not fatal; it is the same status that global phase, absolute gravitational potential, and absolute velocity have throughout physics. The discipline it imposes is that **every physical claim must reduce to a phase difference**. Accordingly, the framework’s entire observable content involving Φ_{bulk} lives in differences between regions — and there is exactly one such observable difference, the gravitational case of §6. Anything said about Φ_{bulk} that does not reduce to a measurable difference is interpretation, and we label it as such.

One distinction must stay sharp, because the revision turns on it: the unobservability of the *absolute* $U(1)$ phase is a gauge fact — only phase differences are physical — and it is independent of the separate question of a preferred Lorentz frame. The framework keeps the former and, as §8 makes explicit, does **not** thereby claim the latter away: the bulk defines a physical rest frame even though its absolute phase is gauge. “Absolute phase is hidden” is not the same statement as “there is no preferred frame,” and we do not use the first to imply the second.

A familiar optical system makes this discipline concrete: holography. A hologram fixes on the plate the interference of an *object* wavefront with a coherent *reference* beam — that is, their *phase difference* (the fringe pattern), never either phase alone. Add a common global phase to object and reference alike and the fringes are unmoved — the gauge fact, “absolute phase is hidden”; advance one *relative* to the other and the fringes shift by a readable amount — the physical difference. (Phase-shifting holography reads out precisely this differential; an off-axis reference contributes only a fixed common carrier that drops out of the reconstructed object, the optical echo of an unobservable uniform background.) One feature of the analogy is load-bearing rather than decorative: **a hologram cannot be recorded in incoherent light**. Without a phase-stable reference there is no stationary fringe — only a washed-out average. This is the optical form of the framework’s central requirement, that a measurement reference be phase-coherent — *one cannot lock to noise* — and it is why Φ_{bulk} cannot be identified with the phase-random zero-point vacuum ($\langle E \rangle = 0$), which belongs instead to the stochastic forcing η of §4.3. We give this as analogy, and label it as such; the frame claim itself rests not on the analogy but on the single global difference of §6.

The coherence requirement also separates three roles the word “vacuum” otherwise blurs. (i) The condensate that locks $L \leftrightarrow R$ and dresses the Dirac mass (§2, §3.6) is a coherent vacuum expectation value — a definite-valued reference. (ii) The measurement reference Φ_{bulk} of this section is likewise coherent and oriented, but its frame-bearing *selection* is the open problem of §8, not a settled value. (iii) The zero-point fluctuations — phase-random, $\langle E \rangle = 0$ — are not a reference at all but the decohering noise of §3.5 and §4.3. Holography forbids collapsing (iii) into (i)–(ii): incoherent light records nothing. The framework’s outstanding obligation is thus confined to (ii) — supplying a coherent, frame-bearing Φ_{bulk} — and §8 carries it as such.

4.3 Magnitude and rate are independent

The smallness of the reference (§4.1) does not weaken the mechanism. The equilibration *rate* — how fast a perturbed system re-locks to the bulk — is set by the coupling to the bath (in MRI, T_1 and T_2), and is independent of the order-parameter *magnitude* (the ppm polarization). A faint reference can drive fast, sharp, definite relaxation. Measurement definiteness rides on the rate, not on the order-parameter size.

This separation has a dynamical companion, and it is where temperature enters as an explicit *randomization term in the phase synchronization itself*. Write the deviation of a locking system from the bulk phase as δ ; near the lock it obeys

$$\dot{\delta} = -K\delta + \eta(t), \quad \langle \eta(t)\eta(t') \rangle = 2D\delta(t-t'), \quad D \propto k_B T,$$

a deterministic restoring force of strength K that pins the axis, plus a stochastic thermal forcing η . The steady state is Gaussian with variance $\langle \delta^2 \rangle = D/K$, and the coherence amplitude of the lock is $r = e^{-\langle \delta^2 \rangle / 2}$. This makes the §4.1 amplitude–axis split precise as a statement about *moments*. Temperature acts through one of them only: it widens the phase *variance* (the second moment) and thereby damps the amplitude r , while the *mean* phase — the axis ψ (the first moment) — is held by the restoring force and does not move. “Temperature randomizes the phase” means exactly this: it inflates $\langle \delta^2 \rangle$, lowering the amplitude, not rotating the axis. The amplitude does depend on the *ratio* D/K , so it is tied to the lock rate; the equilibrium polarization of §4.1 and the axis are the quantities that vary independently of it.

Which fluctuations dominate the forcing η is set by where the relevant transition sits relative to the thermal scale: the thermal occupation of a mode at frequency ω is $\bar{n} = (e^{\hbar\omega/k_B T} - 1)^{-1}$, so for optical transitions ($\hbar\omega \sim 2 \text{ eV} \gg k_B T \sim 0.025 \text{ eV}$) the zero-point (vacuum) fluctuations dominate the selecting noise and the thermal contribution is negligible, while for radio-frequency systems ($\hbar\omega \ll k_B T$, NMR) the forcing is thermal, $D \propto k_B T$ as written above. The noise that selects at an optical photodetector surface is thus vacuum-dominated; at an NMR coil, thermal — one envelope, two regimes of its source.

This thermal half of the envelope is not just a model: it is the measured behaviour of matter-wave interference. Heating C_{70} fullerenes until they emit thermal photons reduces the fringe visibility in quantitative agreement with decoherence theory [28] — temperature damps the amplitude, not the fringe positions, exactly the amplitude-not-axis statement above. The same envelope $e^{-\langle \delta^2 \rangle / 2}$ — the influence-functional form $e^{-\text{Im} \Phi}$ of §3.5 — reappears in §6, where the variance is supplied not by temperature but by a far smaller gravitational phase difference: one slot, two candidate sources — the thermal one measured and standard, the gravitational one contingent on the postulate of §6.2 and absent without it.

4.4 The role of gravity, stated honestly

In ordinary matter the interaction that maintains the bulk reference is electromagnetic — B_0 in MRI, lattice and Coulomb binding in a solid-state detector, supersaturation metastability in a cloud chamber. Gravity is negligible there by ~ 30 orders of magnitude, and we make no claim that gravity maintains everyday detector coherence. In particular, we do **not** assert a gravitational coherence rate of the form $\Gamma_{\text{grav}} \sim GM^2/(\hbar\Delta z)$: summing $N^2/2$ unnormalized pairwise gravitational couplings does not yield a physical relaxation rate (the mean-field coupling must be normalized intensively), and the resulting figure is a dimensional artifact, not a derived rate.

Gravity enters in exactly one place: where the electromagnetic relaxation channel has been deliberately engineered away (the cold, isolated, mesoscopic regime of Penrose–Diósi tests), and, separately, through the ordinary gravitational **redshift** of the electromagnetically-maintained reference clocks at different potentials — which is the basis of the prediction in §6. The first is the classicalization regime Penrose’s objective reduction [15] aims to describe; the second is standard general relativity applied to the detector references, not a new gravitational coherence mechanism.

5. One mechanism, many detectors: capture strength and commit threshold

A single mechanism — capture, selection, registration (§3.1) — spans measurements that look very different. Two numbers characterize a detector architecture along it: the **per-event capture rate** $r \propto \sigma(\omega) |\psi|^2$ (Stage 1 — the energy selectivity lives in the cross-section $\sigma(\omega)$, which *multiplies* the local intensity, so resonance acts as a rate factor rather than a gate), and the **commit threshold** k — how many captures must accumulate at one site, within that site’s memory time τ_{mem} , before Stage 2 commits and Stage 3 can fire. The older unifying axis — how deep inside the Arnold-tongue locking range each capture sits — is the first number; the second is orthogonal to it.

	Stage-1 capture	Reversible window (Stage 2)	Stage-3 registration (reservoir)	Commit threshold k	Reference maintained by
Bell/photon detector	photon resonantly absorbed by a bound electron	brief (resonance fluorescence)	electronic amplification (avalanche, PMT gain; applied voltage)	1	EM lattice binding
NMR / MRI	RF pulse tips M coherently	T2* — refocusable by spin echo	T1/T2 relaxation to lattice	— (ensemble, continuous)	B₀ (applied EM field)
Cloud / bubble chamber	a <i>sequence</i> of weak partial ionizations	partial, per event	stepwise droplet/bubble nucleation (su- persaturation)	1 per vertex, many vertices	EM (metastable su- persaturation)
Photographic emulsion / hologram plate	photolysis in a silver-halide grain	sub-latent- image speck — decays if not reinforced	chemical development (developer reservoir)	≈ 3 –4 per grain [44]	EM (grain chemistry; developer metastability)

Four readings of the same structure:

- **Bell detector — the strong/projective limit.** One event saturates the locking range; the irreversibility is the amplification, not the bare excitation (a single excited electron can re-emit; the avalanche cannot be undone). One quantum, one capture, one commit, one registration, one click. Because $k = 1$ — every capture commits — the per-event statistics are the outcome statistics directly: this is the detector class for which per-event Born claims are properly made.
- **MRI — isolated, deliberately-triggered Stages 2–3.** Here B_0 sets the reference axis and Larmor frequency $\omega_0 = \gamma B_0$ (the role of “ Φ_{bulk} ”); the *gradient* coils are not the reference but the spatial encoder, the direct analog of the Stern–Gerlach gradient that converts internal phase into a position/frequency label. The RF pulse is the coherent capture; relaxation is Stages 2–3, watched directly. Every free-induction decay is, in framework terms, a Stage-2/3 emission measurement — and the ensemble character of the readout means no single-quantum commit threshold applies.
- **Cloud chamber — the weak/repeated limit.** The charged particle undergoes a *sequence* of distinct vertices, each a weak partial capture (removing tens of eV from a MeV–GeV particle) seeding a local commit and nucleation. It is not one drawn-out measurement but many partial capture–commit–register triplets: one particle, many sites — $k = 1$ per vertex, with the number of *vertices* the large number. The collinearity of the track (Mott 1929 [16]) measures the per-event partiality: each capture only partially sharpens the momentum, so the next fires preferentially along the line of flight. dE/dx is the per-event coupling strength integrated over the trajectory.
- **Photographic emulsion — the integrating limit.** Many quanta, one site: a grain commits only after $k \approx 3$ –4 photolysis events build a stable latent-image speck (Gurney–Mott [44]), and a sub-threshold speck *decays* if the next capture does not arrive within its memory time. That decay is **reciprocity failure** — the photographic fact that a total exposure delivered slowly records less than the same exposure delivered fast — and in the present division it is direct, everyday evidence that Stage 2 and Stage 3 are physically distinct: Stage-2 progress can be made and then lost without ever reaching registration. An integrating detector reports a *nonlinear functional* of $|\psi|^2$ (the Hurter–Driffeld response), which is why per-event Born statistics is a $k = 1$ claim, and the emulsion — including the hologram plate — is the wrong exemplar for it.

The same incoming fermion leaves a single click in a silicon pixel and a millimeter trail in a chamber because the two architectures exercise opposite ends of the capture-strength axis — projective saturation versus near-

threshold partial capture — not because the underlying vertex differs. The emulsion adds the orthogonal axis: commit threshold and memory time — one site integrating many quanta, versus one quantum committing at one site.

6. A candidate prediction: a linewidth-dependent gravitational Bell test, and the postulate it requires

6.1 Statement of the candidate effect

The framework *suggests* an observable departure from standard quantum optics — but, as we show in §6.2 and derive in Appendix C, the effect does **not** follow from the framework together with standard quantum electrodynamics. It requires an additional, non-standard postulate. We state the candidate effect here and isolate the postulate it rests on in §6.2; the experiment is best read as a falsification test *of that postulate*, not as a clean consequence of the framework.

Two detectors A and B in a photonic Bell test sit at gravitational potentials $\Phi_A \neq \Phi_B$. Their electromagnetically-maintained reference clocks are redshifted relative to each other by the standard factor $\Delta\Phi/c^2$. *If* the local reference couples into the polarization measurement (the postulate of §6.2), then over a measurement the relative reference phase accumulates

$$\delta\phi_{\text{grav}} = \omega \frac{\Delta\Phi}{c^2} \tau_{\text{coh}} = \frac{\omega \Delta\Phi}{c^2 \Delta\nu},$$

where $\tau_{\text{coh}} = 1/\Delta\nu$ is set by the photon **linewidth**, independent of central energy. The *functional form* $-\cos(2(a-b))$ is set by the polarizer geometry (a reversible basis selection that precedes Stage-1 capture) and is gravitationally invariant. The *contrast* would then degrade as

$$\text{CHSH}(\Delta\nu) = 2\sqrt{2} \exp\left(-\frac{\delta\phi_{\text{grav}}^2}{2}\right)$$

(Gaussian line). Standard QM/QED predicts CHSH independent of linewidth at fixed entanglement fidelity; the candidate effect is a linewidth-dependent degradation in a gravitational gradient. Whether it exists at all hinges on §6.2.

6.2 The postulate the effect requires, and why it does not follow from standard QED

Magnitude, conditional on the coupling. Were the local reference to couple into the projection, the redshift would enter as a *steady, deterministic* frequency offset, not a stochastic per-period kick; a steady offset accumulates phase linearly in time, so $\delta\phi \propto \tau_{\text{coh}} = 1/\Delta\nu$. The contrary “per-optical-period floor” reading, which would make any effect negligible, is incorrect for a steady offset. This fixes the *scaling* of the candidate effect — but says nothing about whether the coupling exists.

It does not follow from standard QED (Appendix C). We attempted to derive the required coupling from the standard light–matter Hamiltonian. The absorption matrix element factorizes into a polarization projection (carrying the analyzer angle) times a temporal/energy factor (carrying the local-clock phase, redshift included). These multiply; they do not mix. The clock phase enters the detection probability only through the modulus of the temporal factor, $|T|^2$, which is independent of the analyzer angle — so the redshift is a common-mode phase that cancels in the normalized correlation. Standard QED therefore predicts CHSH independent of linewidth: the null.

For the effect to exist, the H and V branches must carry different local-clock phases — the local temporal reference must couple to the *polarization projection itself*. We checked every apparatus element that could plausibly supply this, and none does:

Element	Dissipative?	Polarization- dependent phase?	Carries A–B redshift into the correlation?	Why it fails
Detector electron	yes	no	no	H and V absorbed at the same potential $\Phi_{\text{A}} \rightarrow$ common local phase
Polarizing beam splitter	no (unitary)	yes (birefringence)	no	reversible $\rightarrow$ a recalibratable, linewidth-independent phase, not a decoherence envelope
Absorbing polarizer	yes, on the <i>rejected</i> photon	no	no	post-projection the transmitted photon retains no H/V memory; same-potential null
Path deflection / crystal recoil	normally negligible / reversible	yes (recoil)	no	Stern–Gerlach-like and reversible; any recorded recoil references the same crystal at Φ_{A} for both branches $\rightarrow$ common phase

The four failures share one root cause: at a single detector both Bell branches reference the *same* local potential, and every genuinely dissipative step occurs *after* polarization projection, when the photon no longer carries which-branch information. The A–B redshift can enter the correlation only through a coherent, pre-projection, polarization-*differential* phase — and the sole element that fits (birefringence) is unitary, recalibratable, and linewidth-independent.

The required postulate, stated explicitly. The candidate effect therefore rests on an additional term, schematically

$$H' \sim g(\mathbf{d} \cdot \hat{\mathbf{H}}) \phi_{\text{bulk}},$$

coupling the polarization projection directly to the local background phase. This term is (a) absent from standard QED; (b) non-covariant — it singles out the analyzer axis and so breaks the rotational symmetry of the dipole interaction; and (c) tied to the absolute bulk phase, and hence to the physical preferred frame the framework openly carries (§8), whose Lorentz-covariant formulation remains open. We are specific about the defect and its repair: coupling to the *absolute* phase ϕ_{bulk} violates the $U(1)$ /shift symmetry under which only the gradient $\partial_\mu \theta$ of a phase is physical; a gauge-legal form couples instead to that **gradient** — a physical Goldstone current of the bulk order parameter — rather than to the bare phase. The $1/\Delta\nu^2$ scaling survives this repair, for a simple reason: the observable content of the postulate was never the absolute phase but the local *rate* of the reference, and that rate is the temporal component of the gradient, $\partial_t \theta = \omega_{\text{ref}}(\Phi)$ — precisely the quantity gravitational redshift shifts, $\omega_{\text{ref}}(\Phi_A) - \omega_{\text{ref}}(\Phi_B) = \omega \Delta\Phi/c^2$. A gauge-legal coupling $H'' \sim g(\mathbf{d} \cdot \hat{\mathbf{H}}) \partial_t \theta$ therefore still enters as a steady, deterministic, polarization-differential frequency offset between the wings, accumulating phase linearly over $\tau_{\text{coh}} = 1/\Delta\nu$; the $\delta\phi_{\text{grav}} \propto 1/\Delta\nu$ accumulation, and hence the $1/\Delta\nu^2$ visibility exponent of §6.1, carry over unchanged, while only the physically-empty absolute offset drops out. (This is the qualitative argument; a full re-derivation with the gradient coupling is not

carried out here.) What the repair changes is the postulate’s status, not the signature: H'' is shift-symmetric but remains non-covariant — it still singles out the analyzer axis — and absent from QED. Either way the term is presented as a deliberately symmetry-reduced *falsification term*, not a natural consequence of the framework.

What the experiment tests. Because the effect does not follow from the framework plus standard optics, the experiment is a clean falsification test *of the postulate H'* , not of the framework’s core. A null result (CHSH flat versus linewidth) confirms standard QED and disconfirms H' ; a degradation scaling as $1/\Delta\nu^2$, present only with an altitude split, would be evidence for it. We do not claim the effect is a derived consequence of MCI — we have shown it is not — and we present it as the one place the framework *could* be made to touch experiment, contingent on a postulate we cannot currently derive.

Two consistency notes survive the demotion. First, even under H' the clock phase enters only as a multiplicative visibility envelope (a contribution to the $\text{Im } \Phi$ decoherence functional of §3.5, present only under H'); it does **not** generate the correlation. Standard QM supplies both $-\cos 2(a-b)$ and the $2\sqrt{2}$ ceiling — the clock-as-hidden-variable model alone is sub-classical (Appendix A) — so the framework never double-counts the clock as the source of the correlation. Second, the established gravitational-time-dilation visibility effects [24, 25] do not apply here: they require a real internal clock in a spatial superposition straddling both potentials, which the Bell photons lack — which is precisely why an extra postulate is needed rather than an existing mechanism.

6.3 Consistency with existing satellite tests

Existing long-baseline Bell tests do not already constrain the postulate of §6.2. The Micius satellite experiments and ground tests use broad-linewidth spontaneous parametric down-conversion (SPDC) photons ($\Delta\nu$ of order GHz–THz); for those, $\delta\varphi_{\text{grav}}$ is of order 10^{-7} or smaller even over 500 km altitude differences, far below resolvability. The candidate effect is specifically *narrow-linewidth*: it would require cavity-filtered photons ($\Delta\nu \sim \text{kHz}$) to bring $\delta\varphi_{\text{grav}}$ to order unity at km-scale altitude differences. This is why the signature has never been tested — no Bell test has combined narrow linewidth with a gravitational gradient.

6.4 Parameter space and falsification

The discriminating measurement is a joint linewidth $\times$ altitude scan with cavity-filtered narrow-linewidth entangled photons. Photon central energy is not the discriminator; linewidth $\Delta\nu$ and altitude split $\Delta\Phi$ are. Order-of-magnitude: $\delta\varphi_{\text{grav}} \sim 1$ requires roughly $\omega(\text{gh}/c^2)/\Delta\nu \sim 1$, i.e. for optical $\omega \sim 4 \times 10^{15}$ rad/s and a few-km altitude split ($\text{gh}/c^2 \sim 3 \times 10^{-13}$), a linewidth of order kHz. A calibrated scan at fixed entanglement fidelity that holds CHSH flat across linewidth would disconfirm the postulate H' of §6.2 (and confirm standard QED); a degradation scaling as $1/\Delta\nu^2$, present only with an altitude split, would be evidence for it.

A second, independent gravitational handle targets the *ontology* rather than the H' postulate: reading $|\psi|^2$ as a real energy density makes a delocalized mass source a mean field $\propto E|\psi(x)|^2$ (semiclassical, Møller–Rosenfeld), so it would *not* gravitationally entangle two masses — a null in the Bose–Marletto–Vedral tabletop test, where quantized gravity predicts entanglement. This is a far-future test that inherits the known difficulties of semiclassical gravity (it plausibly requires a Diósi–Penrose-type collapse, already constrained by matter-wave and spontaneous-radiation bounds), and is noted here as a discriminator of the wave-energy *reading*, not as a prediction the framework’s core requires.

7. Relation to other interpretations

7.1 An honest placement: the single-world, ψ -ontic, nonlocal family

MCI takes ψ to be a real physical field (ψ -ontic, in the sense of Harrigan and Spekkens [17], consistent with PBR [18]) and adds a beable: the actual configuration of the background/vacuum field at each event, whose value selects the outcome. Its proper category is therefore the **single-world, ψ -ontic, nonlocal**

hidden-variable family — alongside Bohmian mechanics [2, 3] (beable: particle position) and Nelson’s stochastic mechanics [19] (beable: a stochastic background). This is the framework’s real intellectual home, and naming it honestly is more durable than positioning MCI as a slayer of any rival.

The ψ -ontic reading is not a bare metaphysical preference; it is the stance under which several long-established results of single-particle Dirac theory read most naturally. Ohanian [38] recovers the electron’s spin and magnetic moment as the angular momentum of the *circulating energy flow* and the moment of the *circulating charge flow* of the Dirac field — observables obtained by treating the field as a real, spatially-extended distribution rather than a probability amplitude over a point. The Gordon decomposition [39] makes the structure explicit: the conserved Dirac current splits into a convection current and a spin magnetization current, which in the exact Dirac–Coulomb solution of hydrogen [40, 41] are the real circulating currents of a stable, spatially-distributed bound state — the orbital current carrying the orbital moment, the magnetization current carrying the spin moment with $g = 2$ — whose interference reproduces the fine structure, and whose internal circulation sits at the Compton/zitterbewegung scale of §2.4. None of this *proves* field realism — each result also admits an operational reading, and the integrated spin and moment are notably independent of the wave-packet envelope, so the support is for an *extended real field*, not a particle of definite radius. But the field-realist interpretation is the one under which spin, the magnetic moment, and the hydrogen spectrum are properties of a single real, extended object; we take that consonance as independent motivation for the ψ -ontic commitment, not as a derivation of it.

7.2 Against Many-Worlds: a trade, not a conquest

The contrast with Everett [1] is a *trade of costs*, and we state it as such. The moment one insists on a single world with definite outcomes that reproduces quantum statistics, Bell’s theorem forces nonlocality. MWI escapes Bell only by denying definite single outcomes — it buys **locality** at the price of **many worlds**. MCI makes the opposite trade: it buys a **single world** at the price of **nonlocality** (in the Bell sense — not signaling; §7.5). No interpretation pays neither price. So MCI does not *abolish* the nonlocality that troubled Einstein; it relocates the cost from branching worlds to nonlocal beables, as Bohm does. We accordingly describe MCI as *a single-world alternative to Many-Worlds*, not a replacement that escapes its difficulties for free.

7.3 Against Bohm and Nelson: what MCI still owes

Within its own family MCI is currently the least complete member, and honesty requires saying so. Bohm has an explicit dynamical law (the guidance equation) and a clean account of the Born measure (quantum equilibrium, $|\psi|^2$). MCI now adopts the first — the coherent guiding law is the Madelung velocity $\mathbf{v} = \nabla\varphi/m$ of §2.2, read ontologically (§2.5). For the rest MCI makes a different choice than Bohm: rather than *derive* which basin, it **inherits** the selected outcome (§1.3), so its core owes no selection equation. The dynamical law that would fix the outcome — Bohm’s achievement — MCI carries only as a *speculative candidate* (the unwritten background-field selection of §3.3), and its Born account is parked (the fork of §3.3). Its nonlocality is the standard non-separability of the entangled field (§7.5); only the candidate would add a further, superluminal selection nonlocality, which the core does not assert. What this section adds is the comparative point: reading $|\psi|^2$ as the energy density of a real field is true but, for the equal-energy channels the framework treats, coincides with the occupation fraction by normalization and so explains nothing about outcome statistics — leaving MCI without Bohm’s clean quantum-equilibrium answer. (The substrate-sampling model of §8 is the candidate mechanism-level refinement — an equilibrium bath converting a linear real-field drive into $|\psi|^2$ -weighted selection — but it relocates the postulate rather than removing it, so the debt stands.) The companion selection paper [47] has since developed that substrate-sampling germ into a proved fair game — see the §3.3 status update — which, if its premises hold, gives MCI an answer of a *different kind* from Bohm’s: fairness derived from detector physics rather than an equilibrium measure postulated on configuration space. The in-paper debt statement stands for this paper alone.

This recasting also fixes what a superposition *is*, and the point is worth making because the careless version invites a hidden-variable misreading. What is definite is the field: a system “in superposition” is a single definite ψ -configuration that merely fails to be an eigenstate of the observable one chooses — superposition is a relation between the definite field and a chosen reference basis, not an indefiniteness in the field. What

is *not* a property of ψ is the outcome, which is co-produced at Stage 2 by ψ together with the per-run background configuration (§3.3) and fixed only when that background is sampled. This is precisely Bohm’s structure — a definite ψ with a contextual outcome — with the beable changed from particle position to background field, so that “definite” never attaches to the outcome and no local outcome-predetermining variable is smuggled in (Appendix A). The reading earns its place: an observable’s result is background-sensitive exactly when ψ is not its eigenstate, so “superposition with respect to A” is the condition that the Stage-2 selection depends on the background, and the Born weight $|\langle a|\psi\rangle|^2$ is, on this reading, the measure of those backgrounds that select basin a . Why that measure takes the squared-overlap value is the §3.3 fork again, carried in §8.

Against these debts stand three distinctions, and they are sharper than “a more physically-motivated beable.”

First, **the guiding field is physicalized, not posited.** Bohm’s quantum potential $Q = -\hbar^2 \nabla^2 R / 2mR$ is read off the amplitude of ψ after the fact; it has no source of its own and no dynamics independent of the wavefunction it is extracted from. In MCI the analogous guidance is *inherited from the vacuum the particle traverses*: minimal coupling is carried not by the bare particle wavefunction but by the vacuum’s virtual-pair fluctuations, which coherently *dress* the particle’s phase — the reversible $\text{Re } \Sigma$ dressing of §2.5 and §3.6, not a dissipative lock; §2.3 reserves “locking” for the open regime, and in free flight only the closed regime runs (§2.5) — so the particle’s phase is the vacuum phase it tracks. We are precise about which half of Bohm’s amplitude–phase split this captures. The directly physicalized quantity is the *phase* — the Hamilton–Jacobi S — not literally the amplitude-side Q . (That this phase appears as the Aharonov–Bohm phase $e\Phi/\hbar$ recovered from zero-frequency locking is developed in separate work [26], and is not relied on here.) The amplitude side enters through the *same* coupling’s fluctuations: the variance $\langle \delta^2 \rangle$ of the residual phase mismatch about the locked solution damps fringe contrast, and that loss of R -coherence is the decoherence (amplitude) physics. Where Bohm carries two formally separate objects — a guidance equation acting on S and a quantum potential built from R — MCI traces both to one mechanism, the particle’s coupling to the fluctuating vacuum: the coherent dressing supplies the guidance; its fluctuation variance — sampled dissipatively only at boundaries (§2.5) — supplies the amplitude/decoherence side. We present this as a structural correspondence, not an identity $\langle \delta^2 \rangle = Q$.

Second, **it is relativistic from the start.** Textbook Bohm begins from the Schrödinger equation and must then have a preferred foliation grafted on to make its nonlocal dynamics covariant [27] — the standard charge that Bohmian mechanics is non-relativistic. MCI’s Stage-1 dynamics is instead built from Lorentz-covariant Weyl/Dirac objects: the off-diagonal chiral mass coupling $L \leftrightarrow R$, with zitterbewegung as the L – R beat (§2). Covariance is therefore present in the pilot dynamics by construction rather than added afterward. (Single-particle relativistic Bohm already exists in the Bohm–Dirac current-guidance form [27]; MCI’s added content is the Stage-2 synchronization that supplies definite outcomes and the vacuum beable, not the relativistic single-particle law itself.)

Third — the sharpest contrast — **Bohm’s preferred frame is undetectable; MCI’s is physical.** The foliation Bohm needs is, by construction, empirically inaccessible. MCI’s nonlocality is instead anchored to a *physical* preferred frame, the bulk/vacuum rest frame (§8), whose effects are in principle observable. The handle this paper carries is the gravitational channel of §6 — contingent on the §6.2 postulate, and weak, but self-contained: unlike Bohm’s frame, MCI’s is the *kind* of object that can leave an imprint. We take the conservative reading: ordinary matter couples to that frame, beyond the electromagnetic Stage-2 locking, only gravitationally, so the residual anisotropy in everyday particle physics is gravitationally weak (§8). A stronger, *kinematic* coupling to the same frame — a velocity-dependent vacuum-locking envelope, with a fringe-visibility-versus- γ scan as its direct test — is developed in separate work [26] and is *not* relied on here; the in-paper contrast with Bohm stands on the §6 channel alone. Whether that channel suffices is what would decide whether Bohm’s hidden foliation has, in this framework, become a measurable physical frame.

The distinctive debt remains the two items above. The distinctive appeal is now stronger than a slogan about beables: the beable, the guidance, and the locus of nonlocality are one physical thing — the vacuum the particle is locked to — rather than three independent postulates.

7.4 Penrose–Diósi

Both this framework and Penrose–Diósi [15, 20] cast gravity as an agent of classicalization. We localize that role precisely: gravity acts at the electromagnetically-suppressed regime (Stages 2–3 in cold, isolated, mesoscopic systems), never on the basis selection that builds the correlation. Penrose’s objective reduction is a candidate description of exactly that regime; the two proposals single out the same physical step (bulk re-establishment of coherence) and differ in dynamical reading — threshold instability versus continuous rephasing.

Two refinements sharpen that contrast and locate where the readings part company. First, the divergence runs deeper than threshold-versus-continuous: Penrose–Diósi is a *collapse* theory — it modifies Schrödinger evolution and the superposition is objectively destroyed — whereas MCI is no-collapse, the global evolution staying unitary with the coherence scrambled into the bulk (§3.1). Second, the two mechanisms ask opposite things of the environment: MCI’s Stage-2/3 lock is *dissipative* and needs a bath to carry the phase mismatch into a continuum (§3.5), while Penrose’s reduction is *intrinsic* and needs none, being a property the superposed mass distribution carries on its own. In the deep Penrose regime — cold, isolated, mesoscopic, the electromagnetic and thermal channels deliberately removed — that distinction is decisive: the bath MCI would dissipate into has been engineered away, and we do not assert a gravitational dissipation channel to take its place (§4.4). We flag a tension rather than hide it: §3.7 finds that the only coupling able to perform a non-demolition measurement of the chiral channel — and so to complete the Born account — is a coupling to the scalar mass density $\bar{\psi}\psi$, i.e. exactly the gravitational channel this paragraph declines to assert; completing Born would thus narrow the operator-level gap to Penrose–Diósi to the bare collapse-versus-no-collapse distinction. The honest statement is therefore not that MCI subsumes Penrose but that the gravity-dominated regime is exactly where MCI’s dissipative mechanism runs out of bath while Penrose’s intrinsic one does not — which makes that regime the arbiter between them: a continuous, no-collapse reading (a gravitational coupling that damps coherence smoothly, *below* any Penrose threshold) and an objective collapse predict different things there. That the two readings diverge in this regime is the point made here; the continuous branch — a gravitational reduction of the vacuum-locking rate — is pursued in separate work [26], on which nothing above depends.

7.5 Bell, no-signaling, and the “unhidden variable”

We restate plainly: the framework is **not** superdeterministic. Detector settings are free; the shared bulk environment is a *public, setting-independent* variable that Bell’s theorem permits and that does nothing to relax any Bell assumption. There is no faster-than-light signaling — Bell nonlocality is not signaling, and MCI, reproducing standard quantum statistics, inherits the no-signaling theorem exactly as Bohm and ordinary QM do. The bulk reference — a *local macroscopic reference*, to use a less provocative name for the same object — is an “unhidden” variable only in the benign sense: it makes explicit the phase coherence that realistic descriptions of macroscopic detectors already assume. It is not a Bell-relevant hidden variable, and it is not a route around nonlocality.

It is worth being precise about *which* bulk this denies and which it does not, because the same distinction locates the framework’s nonlocality. Two physically different objects travel under the word “bulk,” and only one is the setting-independent common cause just excluded:

- **The local thermal reference** at each detector — the faint, ppm bias of §4 (B_0 in MRI, lattice and Coulomb binding in a solid-state detector). This performs the local Stage-2 provisional commit and Stage-3 registration, and — through its coherence amplitude $r = e^{-\langle\delta^2\rangle/2}$ (§4.3) — the visibility of the outcome. It is established in the common past of both wings and is independent of the analyzer settings, so it is exactly a Bell common-cause variable; by the theorem (and quantitatively by Appendix A, $\text{CHSH} \leq \sqrt{2}$) it *cannot* be the source of the correlations. Its coherence magnitude is irrelevant to this — a more coherent common cause is still bounded by $\text{CHSH} \leq 2$ — so the smallness of the ppm bias is not the obstacle and additional coherence is not the cure. Cooling a detector lowers $D \propto k_B T$ (§4.3), raises r , and thereby sharpens the *visibility* of an entanglement-sourced violation; it does not, and cannot, convert this local reference into a nonlocal channel.
- **The extended, non-separable field configuration** carrying the entangled two-particle mode across

both wings. This is not a common-cause variable at all: it is a genuinely non-separable (entangled) configuration of the ψ -ontic background field, and non-separability is precisely the property a local common cause lacks. *This* is where the framework’s nonlocality resides. Ascribing it here is not a new postulate but a restatement of the ψ -ontic commitment of §7.1 — the wavefunction is a real field, and an entangled pair is one extended, non-factorizable configuration of it spanning A and B.

This nonlocality — the non-separability of the real entangled field — is exactly standard quantum mechanics’, and adds no postulate. It should not be confused with the *superluminal depletion* of the winner-take-all candidate (§3.3): that further, postulated nonlocality belongs to the speculative selection law, which the framework’s core neither needs nor asserts.

Naming the locus this way settles, at the level of *ontology*, the gap §7.3 flagged (“no explicit locus for its nonlocality”): the nonlocal element is the non-separable field configuration, not the local thermal bulk, which does only registration and visibility. It does **not** yet settle the *dynamics* — an explicit law for how that non-separable configuration updates across a spacelike interval (the selection law of §8), together with a proof that the update preserves no-signaling. The preferred frame the framework already carries (§8) supplies the simultaneity such a law needs; the no-signaling proof does not follow for free once the dynamics is written independently of the quantum statistics it must reproduce, and securing it is part of the debt of §8.

7.6 Relational clocks and real-clock decoherence

The framework’s title and its central image — that every particle carries an internal phase clock, and that measurement is the mutual synchronization of such clocks — place it in deliberate proximity to the relational-time tradition, and we state the relationship precisely rather than borrow the vocabulary loosely.

In the **Page–Wootters** mechanism [29, 30] and its modern operational formulations [31–33], time is not a background parameter but a correlation: the global state is stationary, and dynamics is the conditional dependence of a “system” on the reading of a “clock.” MCI uses *clock* in a related but distinct sense. Its clocks are physical phase degrees of freedom — the chiral/Compton phase whose $\pm E$ beat is the Zitterbewegung (§2.4) — and the operation that matters is not the *definition* of a time parameter from a clock reading but the *dissipative phase-locking* of many such clocks to a common reference (§3). MCI does not derive relational time; it takes many physical clocks as given and asks what their entrainment does at a measurement. The two pictures are compatible — a Page–Wootters clock is exactly the kind of physical phase MCI synchronizes — and the quantum-reference-frame treatment of *interacting* clocks [32, 33] is the natural language in which the Stage-1 coupling of a system clock to a detector clock would be made rigorous. We note this as a possible formal home for the framework’s kinematics, not as a result we have established.

Two relational programs bear more directly on the framework’s claims. First, the **thermal time hypothesis** of Connes and Rovelli [34]: for a system in a thermal (KMS) state, the modular flow of that state plays the role of physical time, so a thermal reservoir does not merely *sit at* a temperature but *defines a preferred flow*. This is suggestive for §4, where a faint, overwhelmingly thermal bulk supplies the measurement reference: thermal time offers a candidate first-principles account of why a hot, disordered reservoir can nonetheless fix a definite reference *axis* (the §4.1 amplitude/axis split), and is, we think, the most promising existing framework in which the bulk reference could be grounded dynamically. We flag it as a direction, not a derivation.

Second, and closest of all, is the **Montevideo interpretation** of Gambini, Porto, and Pullin [36, 37], which shares the core intuition that *real clocks produce real loss of coherence*. There, the fundamental imprecision of any physical clock means evolution referred to a real clock obeys a master equation with a universal, clock-induced decoherence rate, so off-diagonal terms decay without a collapse postulate. MCI agrees that decoherence is a physical, clock-mediated process rather than a bookkeeping device, but differs in three specific ways. (i) *Source*. Montevideo’s loss of coherence is fundamental and universal — a property of time itself referred to imperfect clocks — whereas MCI’s is environmental and dissipative: the Adler equilibration of an *opened* system to a bath (§2.3, §3.2), absent for a closed system however imprecise its clock. (ii) *Outcome selection*. Montevideo, like environmental decoherence, yields an improper mixture and does not by itself select a single outcome; MCI adds a beable — the per-run background configuration (§3.3) — to carry the decohered state into one basin. (iii) *Preferred frame*. Montevideo is built to respect general

covariance; MCI openly carries a preferred frame entering through Stage 2 (§8). The honest summary is that MCI is a dissipative, single-world cousin of Montevideo: it relocates the clock-decoherence idea from a fundamental temporal imprecision to a concrete environmental synchronization, then pays for definite outcomes with a nonlocal beable and a preferred frame that Montevideo does not require.

Finally, **relational quantum mechanics** [35] shares MCI’s rejection of any privileged observer but takes the opposite metaphysical route: RQM is perspectival and denies absolute, observer-independent states, whereas MCI is resolutely single-world and ψ -ontic, with one actual field configuration per event. We note the shared anti-anthropocentrism but claim no kinship beyond it.

8. Open problems: what would make this a theory

Two problems would separate a single-world *picture* from a single-world *theory* — but only for the *ambitious* version of the framework that seeks to **derive** outcome selection. The registration account of §3 does not depend on them: it inherits which outcome is selected (§1.3) and stands whether or not these are ever solved. They are recorded here as what the speculative selection candidate (§3.3) would have to supply, together with the routes tried:

1. **The Born measure** — the second horn of the §3.3 fork, which states the gap in full; this entry records only what a derivation must supply, and the routes tried. It must supply *typicality*: the basin geometry is 50/50, flat in $|\alpha|^2$ (§3.3), so the weight has to be carried by the measure on the selecting ensemble; and the equivariance that would carry such a measure intact to the outcome holds only where the Stage-2/3 coupling is non-demolition for the measured observable — which, by §3.7, the electromagnetic coupling is for the which-path basis but not for the chiral channel (there only a gravitational coupling would serve; §7.4). Candidate routes include deriving the relevant symmetry from the U(1) invariance of the chiral phase under global phase redefinition, and the stochastic-mechanics tradition (Nelson [19]); we flag that the latter carries a known obstruction (the Wallstrom single-valuedness condition) and that stochastic electrodynamics has historically reproduced only part of quantum mechanics. A further route is *relaxation to quantum equilibrium*: under a suitably mixing sub-quantum dynamics an arbitrary initial distribution coarse-grains toward $|\psi|^2$, so the Born measure becomes an equilibrium rather than a postulate [42, 43]. This is the natural reading of the framework’s serial background-scrambling — each boundary interaction resamples the unbiased background — but it has been shown for standard Bohmian dynamics, not for MCI’s dissipative, boundary-localized selection, so whether the same H-theorem holds here, and with what coarse-graining, remains open.

A minimal toy model of this route is now in hand (`code/born_substrate_sampling.py`). Threshold-clock detectors driven by a *linear* real coupling in a random-phase (equilibrium) bulk reproduce Born weights by winner-take-all selection. The squared measure arises from the delivered *power*: the sign-alternating linear term averages to zero over the bath phase, and the first surviving term is quadratic — so the route does not smuggle in, through an assumed rate law, the $|\cdot|^2$ it must explain. The same model *breaks* Born predictably away from equilibrium (strong drive, or a coherent bulk), so it carries the non-equilibrium signatures a genuine relaxation account would. What it does not do is prove that MCI’s dissipative, boundary-localized selection *relaxes* to that equilibrium: the bulk’s random-phase condition is assumed, and it is exactly the relocated postulate.

Absent that proof, the $|\psi|^2$ measure is adopted as a **postulate** — Bohm’s quantum-equilibrium hypothesis in this framework’s setting (the second horn of §3.3) — not as a derived result; the relaxation route is a research program, not a discharge of the Born debt.

Status (v8): the companion selection paper [47] now supplies the missing selection process. Four fairness theorems derive drift-free (martingale) share dynamics from detector premises, and optional stopping converts initial energy shares into registration frequencies — Born weights — without an engineered noise ansatz; the noise itself is grounded as the interference cross term between each site’s amplitude and the uncontrolled-phase fields at the detector surface. What remains open moves one level down: the premise identifying energy share with outcome probability awaits a closed microscopic

model, the κ_{ret} ansatz awaits quantization ([47] §9), and the entangled-sector extension consumes an explicit nonlocal shared-registry premise (this paper’s §7.5 locus, given dynamics). The debt recorded above is therefore *relocated into stated premises*, not silently discharged; within this paper the postulate reading stands.

2. **A dynamical law for the selection.** An explicit equation by which the background-field configuration drives the basin selection — Bohm’s analog is the guidance equation. The *ontological* locus of the nonlocality is now identified (§7.5): it is the non-separable field configuration of the entangled pair, not the local thermal reference. What remains is the *dynamical* statement — how that configuration updates across a spacelike interval — together with an explicit proof that the update preserves no-signaling, which is not automatic once the law is written independently of the quantum statistics it must reproduce.

Beyond these, the framework’s relation to Lorentz invariance must be stated openly, because this revision settles it rather than leaving it ambiguous. MCI carries a **real preferred frame**: the rest frame of the bulk/vacuum reference — in the conservative reading, the local mass–energy rest frame. Whether it is instead a *cosmic* frame (the CMB rest frame is the natural candidate) is an open, testable question: only a cosmic frame produces a sidereal signature, and the LIGO sidereal companion test is designed to discriminate the two. We do not pretend the framework is fully relational. What disciplines the commitment is *where* the frame enters. The Stage-1 chiral dynamics is exactly Lorentz covariant; the preferred frame appears only through Stage-2, the dissipative lock to the bulk. We are careful with the terminology: as written — a fixed bulk/vacuum frame entering the Stage-2 interaction — this is **explicit, low-energy Lorentz violation confined to the measurement sector**, *not* spontaneous symmetry breaking. It would become genuinely *spontaneous* only if the preferred frame were the vacuum expectation value of a dynamical field with a Lorentz-covariant action (the Einstein-aether / khronometric construction, with the bulk-phase gradient $\partial_\mu \theta$ as the Goldstone/khronon) — a construction we do not carry out here. Away from measurement, ordinary matter couples to that frame, beyond the electromagnetic locking itself, only **gravitationally** — through the potential differences that produce the redshift channel of §6. Because gravity is weaker than the electromagnetic, strong, and weak interactions that govern particle physics by tens of orders of magnitude, the residual anisotropy this imposes on ordinary, non-gravitational particle interactions is suppressed in proportion — which is why the precise clock-comparison and Michelson–Morley-type bounds have not registered it.

Two caveats keep this honest. First, the *locus* of the vacuum coupling is itself a hypothesis — the **Vacuum Preferred-Frame Hypothesis (VPFH)**: that the quantum vacuum carries a physical rest frame (plausibly the cosmic/CMB frame) and that its coupling to matter is *anisotropic* in that frame. By §2.5 this coupling acts at boundaries, not in free flight, so its signature is a **sidereal modulation of the Stage-2 (boundary) coupling** as the apparatus orientation sweeps past the frame’s axis (the CMB apex) — a second-moment effect, a loss of fringe visibility or squeezing, never a coherent phase shift, which the Michelson–Morley bounds already exclude. Whether the coupling additionally reaches into *coherent, in-flight propagation* — a stronger, velocity-dependent, zitterbewegung-scale form that is **not** gravitationally suppressed — is a further and more aggressive claim. The VPFH is not a commitment of this paper but of the companion analyses (developed separately, and not load-bearing for anything claimed here), which supply its two tests: the LIGO sidereal-anisotropy proposal probes the boundary anisotropy directly, and a fringe-visibility-versus- γ proposal [26] probes the stronger in-flight form (and reads null if the coupling is confined to boundaries). A null in both leaves MCI with only the weak gravitational coupling; a positive result promotes the preferred frame to a boundary anisotropy, and in the in-flight case to a laboratory-velocity effect. Second, a *quantitative* demonstration that the surviving gravitational-strength anisotropy lies below the established preferred-frame bounds is still owed: we have given the reason it should be suppressed, not a proof that it clears every constraint. A fully Lorentz-covariant formulation of the Stage-2 selection — the explicit covariant locus of the nonlocality — therefore remains a third open problem, alongside the Born measure and the dynamical selection law.

This also locates the §6.2 gap precisely. The polarization-dependent coupling of measurement to the local reference, on which the one candidate prediction rests, is a postulate we cannot derive from standard QED (Appendix C) and is non-covariant as written — it is this same preferred-frame commitment, now made

explicit, showing up at the level of a single measurement. On distinguishability: §6 is the one experimental handle, and even it tests that added postulate rather than the framework’s core; the remainder of the framework is, by construction, consistent with standard quantum mechanics rather than distinguishable from it. We state that plainly rather than imply otherwise.

Appendix A: The clock-as-hidden-variable model is sub-classical

This result is the formal justification for confining the framework to the measurement mechanism (§1.3). Consider the simplest local model in which a single classical clock phase per particle determines the click probability by Malus’s law, $P(+1 \mid \hat{n}, \varphi) = \cos^2((\varphi - \theta_{\hat{n}})/2)$. Evaluated for an EPR pair with a shared initial phase (`tests/bell_phase.py`), this returns

$$E_{\text{Malus}}(a, b) = -\frac{1}{2} \cos(a - b), \quad \text{CHSH}_{\text{Malus}} \leq \sqrt{2} \approx 1.414.$$

This is **sub-classical**: it falls below even the local-hidden-variable bound of 2, and far below the quantum value $2\sqrt{2}$. A phase-clock model that treats the clock as a local variable determining the outcome therefore cannot reproduce — cannot even reach the classical bound for — the observed Bell violations. The full spinor (or polarization) structure of standard quantum mechanics is required to recover $-\cos(a - b)$. This is why the framework attributes the correlations to standard quantum mechanics and reserves synchronization for the local measurement mechanism: the alternative is provably ruled out by the framework’s own numerics.

Appendix B: Spin-statistics (consistency check)

MCI does not derive spin-statistics; it inherits it from the underlying Dirac theory and shows where the fermion sign lives. The chiral pair is, as a representation, the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ Lorentz representation; under 2π rotation each Weyl half acquires -1 , and by the standard continuous-deformation argument exchange inherits the same sign. In the Madelung decomposition $\psi_X = \sqrt{\rho_X} e^{i\varphi_X} \chi_X$, the amplitude ρ_X (real, ≥ 0) and the phase φ_X (a real ODE variable, symmetric under particle exchange) carry no sign; only the spinor frame χ_X does. `tests/spin_statistics.py` verifies that the exchange amplitude is -1 to numerical precision across six decades in v/c , and that the Kuramoto phase ODE alone is exchange-symmetric and sign-free. This is a **consistency check** — it confirms the framework’s chiral-pair commitment is compatible with the standard theorem; it could have exposed an internal inconsistency in the reframing but is not, and could not be, an independent derivation.

Appendix C: The redshift cancellation is structural — a Hamiltonian sketch

This appendix supports §6.2: under a standard light–matter Hamiltonian the gravitational redshift enters the polarization correlation as a common-mode phase and cancels, so the candidate effect of §6 requires the additional postulate H' rather than following from the framework plus standard optics.

Setup. A polarization-entangled pair in wavepackets of spectral width $\Delta\nu$ about ω_0 ,

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \int d\omega_A d\omega_B f(\omega_A) f(\omega_B) [|H, \omega_A\rangle |H, \omega_B\rangle + |V, \omega_A\rangle |V, \omega_B\rangle].$$

Each detector is a bulk-bound electron with the standard dipole coupling $H_{\text{int}}^A = -\mathbf{d}_A \cdot \mathbf{E}_A(t)$, referenced to its local potential Φ_{A} .

Factorization. The absorption amplitude for analyzer outcome $|a\rangle = \cos a |H\rangle + \sin a |V\rangle$ separates as

$$M_A(a) = \underbrace{(\varepsilon_a \cdot \mathbf{d}_{eg})}_{\text{polarization projection}} \times \underbrace{\int dt e^{i(\omega_0^A - \omega)t} \dots}_{T_A: \text{temporal/energy factor}}, \quad T_A = |T_A| e^{i\phi_A},$$

where the local-clock phase φ_{A} — carrying the redshift, since $\omega_0 \hat{\text{A}}$ is referenced to Φ_{A} — lives entirely in the temporal factor T_{A} and is *independent of the analyzer angle a* .

Cancellation. The joint amplitude is

$$\langle a, b | \Psi \rangle = \frac{1}{\sqrt{2}} T_A T_B (\cos a \cos b + \sin a \sin b) = \frac{1}{\sqrt{2}} T_A T_B \cos(a - b),$$

so the coincidence probability is

$$P(a, b) \propto |T_A|^2 |T_B|^2 \cos^2(a - b).$$

The phases φ_{A} , φ_{B} — hence $\delta\varphi_{\text{grav}}$ — appear only inside the angle-independent prefactor $|T_A|^2 |T_B|^2$ and drop out of the normalized correlation. CHSH = $2\sqrt{2}$, independent of $\Delta\nu$ and $\Delta\Phi$. This is the null of §6.2.

What would be needed. A surviving effect requires the HH and VV amplitudes to carry *different* phases, $\phi_{HH} \neq \phi_{VV}$ — a polarization-dependent local-clock phase. At a single detector H and V are absorbed at the same potential, so $\phi_A^H = \phi_A^V$ and the requirement fails term by term; the four apparatus elements that might evade this are tabulated in §6.2, and each either references both branches to the same potential or supplies only a unitary (birefringent) phase that is recalibratable and linewidth-independent. The minimal term that would produce the effect, $H' \sim g(\mathbf{d} \cdot \hat{\mathbf{H}}) \phi_{\text{bulk}}$, is absent from QED, breaks the rotational covariance of the dipole interaction, and is tied to absolute φ_{bulk} . The §6 measurement is therefore a test of this postulate, not a consequence of standard theory.

Appendix D: The Stage-2 lock as a conditional measurement

This appendix supports §3.6–§3.7. An earlier version of it derived the Adler/Kuramoto reduction of the Stage-2 lock from an *unconditional* Lindblad equation and read a locked relative-phase as a measurement outcome. **We withdraw that step.** An unconditional dephasing equation yields decoherence, not a single outcome: the averaged density matrix is mechanism-blind — it cannot distinguish “the clocks dephased into the bath” from “one ontic result was realized,” since both give the same $\rho(t)$. The phase reduction $\dot{\phi} = \Omega - K_{\text{eff}} \sin 2\phi$ remains correct where the transverse radius $r \neq 0$, but a fixed point of it is not an outcome; the same equations give $\dot{r} = -2\gamma r \sin^2 \phi$, so the locked direction is generally a *decaying* one, and an unconditional state with σ_x as the measured channel spirals to the maximally mixed center. We therefore replace the argument with the calculation a registration claim actually requires — a conditional quantum trajectory with an explicit monitored record — and are explicit about what it does and does not establish.

Which observable is the pointer. The earlier model monitored the interference channel $\sigma_x = \bar{\psi}\psi$ — the oscillating clock *hand*, which precesses at Ω . Continuously measuring an observable that does **not** commute with the clock Hamiltonian $H = \frac{1}{2}\Omega\sigma_z$ is a *demolition* measurement: the transverse Bloch vector spirals to the origin and the state goes maximally mixed (the “bad zero”). A *faithful, repeatable* measurement instead monitors the **einselected pointer observable**, the one left invariant by the system–pointer coupling [4]: a clock-commuting (QND) observable, here the conserved σ_z (the which-pole basis), not the rotating hand. Aligning the measured observable to the clock axis is therefore not a device to rescue the geometry; it is the standard einselection condition $[H, \sigma_z] = 0$ for a measurement that *records* rather than *erases*. This also exposes an honest limit, consistent with §3.7 and stated again below: ordinary electromagnetic detectors couple to charge/position (σ_x -type) observables, which are demolition couplings for the chiral clock, so how such a coupling einselects a stable σ_z record is left open.

Model. Take the chiral pair as a qubit with clock Hamiltonian $H = \frac{1}{2}\Omega\sigma_z$ (Ω the detuning between the internal clock and the bulk reference). Couple the einselected pointer σ_z dispersively to a damped readout mode a — standing in for the bulk’s collective coordinate — via $H_{\text{int}} = \chi \sigma_z a^\dagger a$, driven to mean occupation $\bar{n}$ and decaying at κ . In the fast-mode regime $\kappa \gg \chi$ the mode is adiabatically eliminated; tracing it out leaves a measurement dissipator in the pointer basis,

$$\dot{\rho} \supset \gamma(\sigma_z \rho \sigma_z - \rho), \quad \gamma = \frac{1}{2}\Gamma_\phi, \quad \Gamma_\phi = \frac{8\chi^2\bar{n}}{\kappa},$$

so the transverse components decay at $2\gamma = \Gamma_\phi$ and the dispersive measurement rate is $K_{\text{eff}} = \gamma = 4\chi^2\bar{n}/\kappa$, the same constant computed from microscopic parameters as before. What changes is its *meaning*: a rate that sets how fast a single run localizes, not a “phase-locking rate” that by itself selects an outcome.

Unconditional state — a mixture of the two pointer states, not the origin. With the QND coupling the Bloch equations are

$$\dot{x} = -\Omega y - 2\gamma x, \quad \dot{y} = \Omega x - 2\gamma y, \quad \dot{z} = 0,$$

so the populations z are *preserved* while the transverse coherence decays. A prepared pure state tilted by θ_0 from the clock axis relaxes to $\rho(\infty) = \text{diag}(|\alpha|^2, |\beta|^2)$ — the interior axis point $(0, 0, \cos\theta_0)$ with $\cos\theta_0 = |\alpha|^2 - |\beta|^2$. This is a *proper mixture of the two pointer states*, not the origin: for a generic prepared state it is partially polarized along the clock axis, not maximally mixed. The unconditional state thus tends to (a mixture of) **exactly the two distinct pointer states** — meeting on its own terms the objection that an unconditional Lindblad dephasing decays to the maximally mixed state — but, being an ensemble average, it still cannot by itself select one.

Conditional trajectory — a single run purifies to a definite, Born-weighted pole. Conditioning on a homodyne record of the σ_z meter (operator $c = \sqrt{\gamma}\sigma_z$, efficiency $\eta = 1$), the state obeys the stochastic master equation

$$d\rho_c = -i[H, \rho_c]dt + \gamma(\sigma_z \rho_c \sigma_z - \rho_c)dt + \sqrt{\gamma}(\sigma_z \rho_c + \rho_c \sigma_z - 2\langle\sigma_z\rangle\rho_c)dW,$$

with photocurrent $dr = 2\sqrt{\gamma}\langle\sigma_z\rangle dt + dW$, where dW is the meter’s vacuum (shot) noise — the physical origin of the trajectory’s stochasticity. For $\eta = 1$ the conditional state **stays pure** and diffuses along the clock axis to a pole $\pm z$: a single monitored run reaches a *definite* outcome at *full purity*. Ensemble-averaging dW returns the unconditional equation above, and the split to the +pole occurs with frequency $|\alpha|^2$ (numerically 0.76 vs the prepared $\cos^2(\theta_0/2) = 0.75$, $N = 400$). The discriminator between “merely dephased” and “outcome realized” is thus the **purity of the conditional state**, not the azimuth: the unconditional average is mixed in every case, while the monitored trajectory is pure and definite — which is why one must track the full ρ , not only the phase ϕ .

Amplification and the cut. A single trajectory is reversible in principle (the catch-and-reverse and spin-echo of §3.2); the outcome becomes a *fact* when the meter signal is amplified above the vacuum noise faster than it can be coherently reversed — a signal-to-noise condition that on the dispersive model requires readout power $\bar{n} \gtrsim \bar{n}_{\text{crit}} = \kappa|\Omega|/4\chi^2$. The Heisenberg cut of §3.5 is this amplification threshold — the Stage 2 $\rightarrow$ Stage 3 boundary of §3.1: below it the interaction is an echo-reversible dephasing; above it the record is irreversibly committed to the bulk. (We do not here re-assert specific critical exponents for that threshold.)

What this establishes, and what it does not. *Established* (standard open-system / continuous-measurement theory [4]): a faithful, einselected conditional measurement of the clock-aligned pointer purifies a single run to a definite pole, while the unconditional average is a mixture of the two pointer states. The appendix therefore does *not* convert a decaying Bloch direction into an outcome — the conditional trajectory, not the unconditional average, is what carries the outcome.

Not established, and flagged:

1. **Born is consistent, not derived.** The across-run weights come out $|\alpha|^2$ because the Born weighting is *built into* the standard measurement unraveling (the dW couples to $\langle\sigma_z\rangle$). The appendix exhibits *Born-consistency*; deriving $|\alpha|^2$ from the dynamics is the second horn of §3.3, left open in §8. The place a derivation would have to act is the amplitude-weighting of the boundary-noise coupling — the route taken by objective-collapse (CSL-type) models — which is not attempted here. (A separate

candidate for the *weight* — that it is the equilibrium statistics of a random-phase bulk, §8, illustrated by `code/born_substrate_sampling.py` — likewise supplies a mechanism given an *assumed* equilibrium, not a derivation.) *v8*: the companion [47] has since carried exactly this step through: its Theorem 1 derives the amplitude-linear noise coupling as the unique drift-free scaling, and grounds it physically as the interference cross term between each site’s amplitude and the uncontrolled-phase surface fields — see the §3.3 status update. The consistency exhibited here is thereby upgraded, in the companion and conditional on its premises, to a derived fairness.

2. **Single-run selection.** In the unraveling used here, *which* pole a given run reaches is fixed by the meter’s vacuum noise (genuine stochasticity). The framework’s alternative — that the outcome is fixed by the per-run background-field configuration (§3.3, §4), the non-separable configuration of the entangled pair (§7.5) — is the hidden-variable completion whose explicit dynamical law is open problem 2 of §8; as a *local* reading it is constrained by Bell (§7.5). The two readings (vacuum-stochastic vs configuration-determined) are not reconciled here. *v8*: the companion’s noise grounding [47] narrows the gap: the meter’s dW is physically identified as the interference cross term between the site amplitude and the uncontrolled-phase fields of the detector’s own charges and the vacuum — a concrete, per-run field configuration — so the vacuum-stochastic reading acquires a configuration-level carrier. A full reconciliation (the explicit selection dynamics, open problem 2 of §8) remains open.
3. **The detector bridge (§3.7).** Whether ordinary electromagnetic detectors — which monitor charge/position rather than $\psi\psi$ — einselect the clock-aligned σ_z pointer at all is the bridge §3.7 leaves open. The model here is the idealized faithful-measurement limit, not a demonstration that real detectors realize it. For the charge/which-path basis that ordinary detectors *do* einselect (§3.7), a resonant charge-coupled readout reproduces the Born weights $|\psi|^2$ energy-independently (`code/detector_resonance_selection.py`); the open bridge is specifically to the chiral σ_z pointer, not to which-path registration.

Reproducibility. The full density-matrix integration — unconditional master equation, conditional homodyne trajectories, and the purity and Born checks — and the figure are produced by the accompanying script `resolution/precession_radius_two_zeros.py` (numpy + matplotlib, seed 7). The tests check the complete ρ (Bloch length / purity), not only the extracted phase angle.

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